Pricing Frictions, Tax Incidence and Welfare: Evidence and Theory

Yu-Chun Elisa Cheng*† Yeliz Kacamak ‡ Eleanor Wilking§

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Abstract

This paper explores how optimization frictions in supplier price-setting affect tax incidence. We use a natural experiment from the staggered adoption of Voluntary Collection Agreements (VCAs) by a major rental platform, which created an effective tax increase for hosts. On average, the tax-inclusive price paid by consumers rose by roughly 5.4 percent, implying that hosts absorbed about half of the statutory tax by cutting their listed prices. However, this average effect masks significant heterogeneity. Pass-through is lowest for sophisticated hosts—defined as those whose price movements before the policy closely follow local hotels—who pass through only about a quarter of the tax, while high-friction hosts pass through roughly two-thirds. We develop a model with heterogeneous price-setting frictions that shows how unsophisticated hosts "anchor" the market price to the previous equilibrium, thus shifting the tax burden to consumers. Counter-intuitively, this mechanism can lead to higher post-tax profits for all firms compared to a market without frictions.

Keywords: tax incidence, pricing frictions, tax compliance, salience

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*Authors listed in alphabetical order.

†Cornell University; yc2572@cornell.edu

‡Boğaziçi University; yeliz.kacamak@bogazici.edu.

§Cornell University; ewilking@cornell.edu.

1 Introduction

In classic models, tax incidence—the economic burden of a tax—is fully determined by relative demand and supply elasticities (Myles, 1989; Weyl and Fabinger, 2013). Newer models, influenced by behavioral economics, reflect that individual consumers are influenced by a multitude of optimization frictions that affect relative tax burden in practice (Chetty, 2009; Finkelstein, 2009). However, in certain markets, *suppliers*—individuals with minimal price-setting experience—are affected by these same optimization frictions. How does this shape tax incidence and welfare?

We explore this question in the context of short-term rental markets using listings data from a major platform firm, Airbnb. Several American cities signed Voluntary Collection Agreements (VCAs) at different times between 2014 and 2016. Per these agreements, Airbnb collected hotel tax at the time of booking on behalf of hosts. Because few hosts remitted taxes prior to VCA implementation and nearly all complied afterward, the agreements resulted in a plausibly exogenous effective tax increase.

We have two main empirical findings. First, using daily property-level price data as well as commercial hotel data, we document substantial pricing frictions among suppliers in the Airbnb market: nearly half of all individual hosts adjust prices infrequently and show limited responsiveness to local demand conditions or competitor hotel rates. In addition, there is evidence that hosts employ heuristic pricing; greater than a third of all hosts list integer prices divisible by five. These rigidities stand in contrast to the more dynamic and flexible pricing strategies employed by hotels.

Second, we exploit plausibly exogenous variation in the specific timing of VCA adoption to measure the pass-through of the effective tax. We use a difference-in-differences framework with estimators robust to staggered treatment timing and heterogeneous treatment effects (Sun and Abraham, 2021; Callaway and Sant'Anna, 2021) to estimate the causal effect of VCA adoption on tax-inclusive consumer prices and demand. The property-weighted average statutory tax rate change across treated cities is about 12 percentage points, so the benchmark for full pass-through is a 12 percent increase in the tax-inclusive price. We estimate that VCA adoption raises the tax-inclusive price by roughly 5.4 percent on average, implying that hosts absorb roughly half of the statutory tax by cutting their pre-tax listed prices. This magnitude roughly comports with Bibler et al. (2021), who use a similar dataset to infer levels of tax evasion among hosts prior to VCA adoption.

However, the more significant discovery lies in the heterogeneity of the effect by our proxy for pricing friction. For each property listing in the period preceding the VCA, we generate a correlation between that property's pricing and local hotel pricing. We find that adjustments varied sharply: "low friction" hosts, whose price movements closely followed those of hotels, absorbed the bulk of the tax, with consumer tax-inclusive prices rising by only about 3.1 percent—roughly a quarter of the full statutory rate—and the estimate is not statistically distinguishable from zero in our preferred specification. "High friction" hosts, in contrast, raised consumer tax-inclusive prices by about 7.9 percent, implying that they passed through roughly two-thirds of the tax. These heterogeneous pricing responses translated into directionally similar effects on bookings, with suggestive evidence that reservation ratios fall more for high-friction hosts—though the estimated effects on reservations are considerably less precise than the price effects.

Taken together, our results suggest that pass-through in markets with significant populations of suppliers who are inattentive, inappropriately anchored, or face other optimization frictions is likely to differ meaningfully from markets where all or most suppliers are traditional firms. This characterization is likely to apply to most amateur suppliers on digital platforms, but could also potentially apply beyond platforms to any context where there is variation in price setting acumen, such as a farmers' market or owner-operated retail. Although there are various studies that establish that consumers face optimization frictions that affect their responsiveness to changes in tax rate or tax administration (e.g., Chetty et al., 2009; Goldin and Homonoff, 2013; Homonoff, 2018; Lockwood and Taubinsky, 2017), there is comparatively little evidence to date on whether similar optimization frictions also affect suppliers. Our findings contribute to an emerging strand within the incidence literature that asserts that the nature of the firm is an underexplored determinant of the burden passed on to consumers (e.g., Kopczuk et al., 2016; Harju et al., 2015).

To more formally understand the welfare consequences of supplier salience and other optimization frictions, we build a generalized model of firm pricing behavior. Building on Anderson et al.'s (2001) work on tax incidence with strategic complementarities in differentiated-product oligopoly, we adapt the Calvo (1983) pricing framework to explicitly model heterogeneous price-setting frictions across firms in an ad-valorem tax setting. Our model generates three novel predictions. First, rigid price-setters create an "anchoring effect" that helps hosts maintain a high market price and shifts tax incidence toward consumers. This mechanism is distinct from the elasticity-based determinants in Weyl and Fabinger (2013). Second, we establish that the existence of frictions give rise to more dead weight loss as well as a counterintuitive welfare ranking in which firms in frictional markets earn higher profits post-tax than in frictionless benchmarks, despite greater market inefficiency, complementing insights on overshifting (Conlon and Rao, 2020) while providing a new mechanism based on heterogeneous price rigidity. Third, we predict that the consumer tax burden is maximized immediately upon implementation and gradually shifts to producers over time, offering testable implications for understanding incomplete and time-varying tax pass-through.

A caveat is warranted. Price changes provide direct evidence of the increased cost of maintaining consumption after the policy, and can also provide indirect insight into underlying market functions (Kopczuk et al. (2016); Stolper (2016)). However, we proceed with caution in inferring that the policy changed the tax incidence, at least incidence in the way that it is conventionally defined—as the ratio of reductions of total consumer and producer surplus that results from imposition of a tax. We interpret our results to suggest that the tax burden on guests in these cities rose—they are now paying higher prices for identical products. Yet, if some hosts were previously evading their remittance obligation, as seems likely, their absolute tax burden rose as well (from zero). Therefore, the policy changed the tax incidence in the sense that it shifted the burden of the tax from taxpayers to the suppliers and consumers, rather than the relative burden of the tax as shared between consumers and suppliers.

Another limitation of our approach is that it relies in the main on data from a single platform firm, in a single industry. While we duly acknowledge that this inherently limits generalizability of our estimates, we maintain that two key features of this context expand the project beyond a case study: first, despite obvious difficulty in valuation, hosts have full autonomy in price-setting, and second, a large contingent of hosts on Airbnb are amateurs—lots of amateurs in the market characterizes other emerging platform or "market-maker" driven markets.

For clarity, we explicitly define key terms employed throughout the paper as follows. We understand pass-through—distinct from incidence—as the degree to which tax exclusive prices adjust to shift the economic burden of the tax to non-remitting parties to the taxable transaction. As is standard, we express pass-through as a percentage calibrated to the total tax liability. We refer to individual suppliers, who list their property on the Airbnb as hosts, and consumers or short-term renters as guests, in keeping with Airbnb's nomenclature.

2 Policy Background and Data

This section provides background information relevant for subsequent analysis. We describe characteristics of the emerging, platform-driven, short-term rental market, and Airbnb specifically. Next, we discuss a timeline of the Airbnb VCAs, which provide the plausibly exogenous policy variation needed for analysis. Finally, we discuss our two main sources of data and key variables.

2.1 Policy

Airbnb is the largest of several firms facilitating short-term, peer-to-peer residential space rentals through an online platform. Its transactions often operate outside the regulatory frameworks governing traditional hotels. Originally conceived as an online marketplace to connect couch surfers, Airbnb has experienced remarkable growth in recent years, expanding exponentially in popular tourism cities around the globe. Hosts on Airbnb create listings for each of their properties. Each listing includes information about the space's characteristics, such as the number of beds, kitchen availability, and whether it is a private apartment or a shared space. Hosts can designate a listing's availability and set its price for each calendar day.

In addition to consumer safety concerns, local governments expressed frustration with Airbnb hosts' avoidance of short-term rental taxes. In cities with significant tourism, the estimated loss of occupancy tax revenue was significant. Initially, Airbnb's position was that its rentals were not subject to occupancy taxes because transactions were "peer-to-peer" rather than commercial in nature. In May 2014, the company officially retracted this view and announced that it believed its hosts were responsible for paying occupancy taxes to local governments. It also amended its "Terms of Service" agreement to inform hosts of their obligation to research and comply with applicable local tax and regulations.

However, there are several reasons that hosts remained unlikely to independently remit tax liability. First, although Airbnb announced that hosts were liable, it is likely that many hosts remained unaware of their obligation. Second, for most cities, occupancy tax remittance systems were designed to conform with professional accounting systems and agents of large hotels, imposing a large compliance cost on individuals. For example, Chicago's tax authority initially did not accept online payment. Instead, hosts were required to complete a multi-page form, print it and and send it along with a paper check each month.

In response to building political pressure, Airbnb began signing Voluntary Collection Agreements (VCAs) with local jurisdictions starting in 2014. Under these agreements, Airbnb collects local occupancy taxes at the point of booking and remits them directly to city governments, effectively shifting the compliance burden from individual hosts to the platform. In return, cities often granted concessions, such as waivers of back taxes, limits on audit access to host-level data, or legalization of short-term rentals under relaxed permitting standards. Between 2014 and 2016, Airbnb signed VCAs with over twenty cities, including several in our sample: Philadelphia, San Diego, Phoenix, Los Angeles, and Boulder. This staggered timing of adoption across cities provides the identifying variation for our empirical strategy.

Because few hosts remitted taxes prior to VCA implementation and nearly all complied afterward, the agreements resulted in a plausibly exogenous effective tax increase for hosts. Crucially, the VCA did not change the statutory tax rate—it changed enforcement.

2.2 Data

Our analysis draws on two primary data sources, covering the period from December 2014 to August 2016. The main dataset is Airbnb listing-level data scraped directly from their website and supplemnted with data from AirDNA, a third-party provider that collects and curates publicly available information from the Airbnb platform. These data offer high-frequency, property-level detail on nightly prices, availability, reservation activity, and listing characteristics across selected U.S. cities and their surrounding metropolitan areas.

To benchmark trends in the traditional accommodation sector, we supplement the Airbnb data with daily market-level hotel data from Smith Travel Research (STR). The STR dataset includes key hotel performance metrics—such as occupancy rates, average daily rates (ADR), and revenue per available room (RevPAR)—aggregated at the city or MSA level. These data enable direct comparisons between digital platform rental activity and hotel market outcomes over the same period.

We focus on a balanced sample of ten U.S. cities, summarized in Table A1. Five adopted VCAs during the sample period: Phoenix, San Diego, Boulder, Philadelphia, and Los Angeles. Five did not adopt VCAs during the sample period: Austin, Dallas, Houston, New Orleans, and Savannah. These cities were selected based on data availability, consistent short-term rental activity, and comparable tourism intensity. The restriction to a smaller, well-observed set of cities ensures consistent data coverage across treatment cohorts and facilitates a clean event-study design exploiting the staggered timing of VCA adoption.

To ensure that our sample captures active short-term rental properties, we impose several restrictions on the raw Airbnb data. First, we exclude extreme outliers in property size by dropping listings with more than five bedrooms or a maximum guest capacity exceeding twelve. These rare listings likely serve specialized or commercial functions and are not representative of typical Airbnb offerings. Second, we restrict the sample to active hosts by requiring that each listing have at least one reservation in the pre-policy period. This step removes approximately 30% of the original sample and ensures we focus on properties with genuine market activity. Finally, we limit the analysis to a short-run window around VCA implementation to isolate immediate behavioral responses. Specifically, we keep observations within a window spanning five weeks before and twelve weeks after the policy's effective date.

3 Empirical Evidence of Host Pricing Frictions

In this section, we present descriptive evidence suggesting that Airbnb hosts face significant frictions in setting pre-tax prices. This evidence motivates our heterogeneity analysis in the next section.

Airbnb hosts differ significantly from hotels and each other in several dimensions relevant to price setting: information access, experience, technologies, cognitive constraints, and adjustment costs. Large hotel chains frequently employ centralized pricing, optimized in corporate headquarters with advanced dynamic pricing algorithms that automatically adjust prices in response to minute changes in local demand. In contrast, most Airbnb hosts have minimal access to data relevant to price setting for their market, have little previous experience in pricing within the hospitality industry, and must manually change their default price for each date using the Airbnb Calendar.[1]

An emerging literature within industrial organization has sought to categorize pricing frictions in peer market pricing and identify their underlying mechanisms (Huang, 2022; Pan and Wang, 2019). For our purposes, however, we aim to establish the existence and extent of pricing frictions; we remain agnostic as to the precise mechanisms that give rise to the market-level pricing patterns we document.

Fact #1. Airbnb hosts rely on pricing heuristics. We first consider a "static" price-setting friction: one that affects the selection of nominal price values. Panel (a) of Figure 1 plots the distribution of nightly listed prices below \$260. The mass of prices is overwhelmingly concentrated on a sparse grid: the largest spikes occur at multiples of ten (e.g., \$100 and \$150, each with roughly 900 observations), followed closely by spikes at multiples of five but not ten (e.g., \$75 with about 900 observations, and \$125 with about 770), while prices ending in any of the remaining digits (shown in light gray) account for only a small fraction of the distribution. It is implausible that optimal prices reliably land on this five-dollar grid, which instead suggests that hosts are using simple heuristics akin to a rounding rule. The figure also reveals smaller, but non-trivial, bunching at digits ending in nine (e.g., \$79, \$89, \$99), indicating that some hosts do attempt to exploit consumer left-digit bias, though this charm-pricing behavior is much less prevalent than rounding to multiples of five.

Panel (b) of Figure 1 reinforces this picture by reporting, for each origin digit (rows), the percentage of price changes that land on each destination digit (columns). The pattern is striking: when a host's current price ends in 0, 43.5% of subsequent revisions still end in 0

[1] During our study period, Airbnb provided very little guidance to hosts in initially setting prices, and no guidance on dynamic price adjustments because of concerns that providing such assistance might alter the legal characterization of hosts as independent contractors, thus exposing Airbnb to liability.

and another 32.1% move to a price ending in 5—together, more than three-quarters of changes from a round price stay on the five-dollar grid. The same is true in reverse: from a price ending in 5, 33.9% of changes return to a 5-ending price and 29.6% move to a 0-ending price. Prices ending in 9 are also highly sticky, with 50.8% of revisions from a 9-ending price ending in 9 again, consistent with the smaller charm-pricing cluster visible in Panel (a). By contrast, transitions originating from non-rounded digits are diffuse and account for only a small share of all revisions. Taken together, the two panels suggest that the typical Airbnb host's pricing strategy is constrained to a small menu of focal values—either the five-dollar grid or, less commonly, just-below-round charm prices—rather than the continuous adjustment one would expect from frictionless dynamic pricing.

Fact #2. Most Airbnb hosts do not engage in dynamic pricing, or use simplified strategies. As tax incidence is determined by adjustments to the pre-tax price, frictions in price changes are highly relevant to our question. Figure 2 plots daily listed prices (demeaned at the property level) over the ten weeks prior to VCA adoption for sixteen randomly selected properties. The panels reveal substantial heterogeneity in host price-setting behavior. Panels 3, 10, and 11 are perfectly horizontal, indicating that the host did not adjust the price at all during the pre-period. Panels 5 and 6 are nearly flat, with only small and infrequent deviations from the mean. Panel 9 displays a simplified dynamic pricing strategy in which the price oscillates between two values, one for weekdays and the other for weekends. Panels 13 and 15 display a coarse step-like pattern in which the price toggles between two values for extended stretches, consistent with simplified "regime" pricing. Panels 4, 7, 14, and 16 oscillate within a narrow band, suggestive of either weekday/weekend pricing or some other low-frequency rule. Finally, panels 1, 2, 8, and 12 exhibit pricing behavior more closely resembling what one might expect from a hotel with access to local demand information, with frequent and large adjustments throughout the pre-period.

In fact, Airbnb hosts vary dramatically in the degree to which they face dynamic pricing frictions. Figure 3 introduces our preferred measure of pricing frictions; for each property, we calculate the correlation between average local daily hotel price series and the property's price series. This measure has several desirable attributes. Unlike other proxies for frictions, such as the number of properties a host has, location, or duration of time on the platform, this measure is both dynamic and continuous, allowing for finer distinctions to be drawn between hosts. In addition, unlike other measures, it directly captures price setting acumen, if we make the reasonable assumption that hotel price movements represent profit maximizing adjustments.

Fact #3. The hotel-price correlation captures meaningful pricing-strategy heterogeneity. Before using the hotel-price correlation as our preferred measure of pricing friction, we verify that it is picking up genuine differences in host price-setting behavior rather than statistical noise. Figure 4 plots, for each property in our analysis sample, the pre-policy correlation between the property's listed price and local hotel prices against the number of distinct prices the host posted during the pre-VCA period. Two patterns are apparent. First, a large mass of properties posts only a handful of distinct prices over the entire pre-period (with many posting just one or two), reinforcing the impression from Figure 2 that most hosts engage in little active repricing. Second, the distribution of correlations forms a tight triangular envelope: hosts with very few distinct prices mechanically cannot attain large positive or large negative correlations, while only hosts who actively reprice can reach correlations approaching ± 1. The hotel-price correlation thus inherits its information content directly from observed repricing intensity—high (in absolute value) correlations are achievable only when hosts actually move their prices—and is therefore a meaningful summary of how actively a host responds to underlying demand conditions, rather than an artifact of measurement noise. We interpret this as supporting the use of the correlation as a continuous proxy for pricing friction in our subsequent analysis.

Summary statistics. Table 1 reports descriptive statistics separately for the pooled analysis sample and for high- and low-friction listings, using the pricing friction measure described above. Panel A reports means, standard deviations, and medians for 5,658 properties in treated cities, split into 3,159 high-friction and 2,499 low-friction listings based on whether their pre-policy hotel-price correlation falls below or above the sample median. Panel B reports analogous statistics for 3,465 properties in control cities. Although average listed prices in treated cities ($126) are lower than in control cities ($166), the distributions are sufficiently noisy that we cannot rule out that the means are the same. Similarly, within treated cities, high-friction listings have slightly higher listed prices on average ($131 vs. $120) and have modestly higher reservation ratios (0.66 vs. 0.59) than low-friction hosts, but we cannot rule out that these differences reflect noise. Tax rate changes are balanced across friction groups within treated cities, with an average rate change of approximately 12 percentage points, confirming that the heterogeneous responses we document in the next section are driven by differences in host behavior rather than differences in tax exposure.

4 Empirical Analysis

Throughout this section, we distinguish between two price concepts. The *pre-tax listed price* (or simply "listed price") is the nightly price advertised by the host on Airbnb, exclusive of any occupancy taxes. The *tax-inclusive price* is the total price paid by the consumer, which includes the listed price plus any applicable taxes. Prior to VCA adoption, few hosts independently remitted occupancy taxes, so the tax-inclusive price was effectively equal to the listed price. After VCA adoption, Airbnb began collecting and remitting the tax on top of the listed price, driving a wedge between the two. A host who does not adjust their listed price in response to the VCA therefore passes the full tax through to consumers.

4.1 The Effect of VCA on Host Price Adjustment

To identify the effect of VCA adoption on host pricing, we use difference-in-differences (DiD). The regression equation takes the following form:

$$log(P_{ijt}) = \beta \cdot VCA_{jt} + \lambda_i + X_{jt} + \varepsilon_{ijt} \tag{1}$$

where P_{ijt} is the log tax-inclusive price for property i in city j at year-week t—that is, the total nightly price paid by the consumer, equal to the host's listed price plus any applicable occupancy taxes collected and remitted under the VCA. We focus on the tax-inclusive price because it is the price consumers actually face and therefore the object that determines tax incidence: a host who fully absorbs the tax by cutting their listed price produces no change in the tax-inclusive price, while a host who leaves their listed price unchanged passes the full statutory rate through. The main independent variable VCA_{jt} is a dummy variable equal to 1 if the VCA agreement is in effect in city j at time t, and 0 otherwise. We include property fixed effects λ_i to control for observed and unobserved time-invariant property-specific characteristics, and metro-week covariates X_{jt} to capture metro-specific seasonal demand shocks. Finally, ε_{ijt} is the error term. All estimations are clustered at the MSA level. Because VCA adoption is staggered across cities, standard TWFE estimates may be biased due to heterogeneous treatment effects (Goodman-Bacon, 2021). We therefore also report estimates using the Callaway and Sant'Anna (2021) estimator, which is designed for staggered adoption settings, using never-treated cities as controls.

Identification rests on three assumptions. First, the *parallel trends* assumption requires that, absent VCA adoption, outcomes in treated and control cities would have evolved along similar trajectories. We probe this assumption directly in the event study specifications below and find no evidence of differential pre-trends. Second, the *no-anticipation* assumption requires that hosts did not adjust their pricing behavior in advance of VCA implementation. This is plausible in our setting because VCA agreements were typically negotiated between Airbnb and local tax authorities with little advance public notice to individual hosts. Third, we require that VCA adoption in one city did not affect pricing or reservation outcomes in control cities (*no spillovers*). Given that short-term rental markets are highly local, cross-city spillovers are unlikely over the short time horizon we study.

Table 2 reports the results for the log tax-inclusive price. Column (1) provides a basic DiD estimate from a two-way fixed effects model, and Column (2) adds city-specific year-week demand controls to capture seasonal changes. Columns (3) and (4) use the Callaway and Sant'Anna (2021) estimator, which is robust to heterogeneous treatment effects in staggered adoption settings. Column (4), which includes demand controls, is our preferred specification. The point estimates are positive across all specifications, indicating that the price consumers pay rises when a VCA takes effect. Our preferred estimate (Panel A, Column (4)) implies that the tax-inclusive price rises by roughly 5.4 log points after VCA adoption (equivalent to about a 5.5% increase).[2] The property-weighted average statutory tax rate across treated cities is 12.3%, so the relevant benchmark for full pass-through in log points is $\log(1.123) \approx 0.116$ (equivalent to a 12.3% increase in the tax-inclusive price). The pooled estimate therefore implies that hosts

[2]Throughout this section we report coefficients in log points because the DiD coefficient on $\log P$ is, by construction, a log-point effect; for the small magnitudes we estimate, the percent equivalent is $100 \cdot (e^{\beta} - 1) \approx \beta$.

absorb roughly half of the statutory tax by cutting their listed price, while the remaining half is passed through to consumers.[3]

Panels B and C explore heterogeneity in this response by our proxy for pricing friction. Hosts whose pre-policy price movements were less correlated with local hotel prices—our "high friction" group (bottom 50% of price correlation)—exhibit a tax-inclusive price increase of about 7.9 percent, implying that roughly two-thirds of the statutory tax is passed through to consumers. In contrast, for "low friction" hosts (top 50% of price correlation), the tax-inclusive price rises by only about 3.1 percent—roughly one-quarter of the full statutory rate—and the 95% confidence interval includes zero, so we cannot reject that low-friction hosts fully absorb the tax. The contrast between Panels B and C is consistent with the theoretical prediction: hosts with greater pricing frictions fail to adjust their listed price in response to the VCA and therefore pass more of the tax through to consumers, while low-friction hosts behave closer to the frictionless benchmark and absorb the bulk of the tax.

Acknowledging that using the median correlation as a threshold to classify high and low pricing friction is somewhat arbitrary, we investigate the heterogeneous effects of VCA reform across listings at different points of the price correlation distribution. More specifically, we re-estimate the staggered DiD specification separately for listings in each decile of the pre-policy hotel-price correlation. Figure 5 reports the resulting estimates of the ATT on the log tax-inclusive price, with each point on the horizontal axis corresponding to a decile of the correlation distribution (decile 1 = the most friction-constrained hosts, decile 10 = the least). The estimated ATTs are uniformly positive but decline monotonically across deciles, from roughly 8.5 percent in decile 1 to a precisely-estimated zero in decile 10. This pattern is consistent with the theoretical prediction: hosts with higher correlations to local hotel prices—and thus lower pricing frictions—reduce their pre-tax listed price by more, absorbing a larger share of the tax and limiting the increase in the tax-inclusive price faced by consumers, while hosts in the lowest correlation deciles pass the tax through almost fully.

Having established how hosts adjust their pre-tax listed prices in response to VCA adoption, we now turn to how these adjustments appeared from the consumer's perspective. Because the

[3] The property-weighted average is pulled up by Los Angeles, which accounts for roughly two-thirds of the treated properties and carries a 14% statutory rate; the simple (unweighted) mean across the five treated cities is 9.1%.

VCA introduces a tax collected on top of the listed price, the net effect on the price consumers actually pay depends on the extent to which hosts offset the tax by lowering their listed price. To capture this and to trace its dynamic evolution, we estimate an event study specification in which the dependent variable is the log tax-inclusive price—the total price paid by the consumer, including the listed price plus any applicable occupancy taxes:

$$log(Y_{ijt}) = \sum_{k=-5}^{k=12} \beta_k \cdot VCA_j \times \mathbf{1}[t = c + k] + X_{jt}\psi + \epsilon_{ijt}. \tag{2}$$

where $\mathbf{1}(t = c + k)$ is a dummy variable indicating the number of relative weeks to the VCA reform, where k does not include -1 since we use the week before the reform as the reference period. β_k captures the dynamic effects of the VCA reform on the outcome relative to the reference period.

Figure 6 displays the estimated β_k from Equation (2) using our preferred estimator from Callaway and Sant'Anna (2021). The horizontal axis represents the number of weeks relative to VCA implementation, and the bars indicate 95% confidence intervals. Black circles denote the pooled estimate across all properties, blue triangles denote high-friction properties (bottom 50% of price correlation), and orange diamonds denote low-friction properties (top 50%).

Two key insights emerge from the dynamic DiD analysis. First, there is no evidence of a pre-existing trend: the coefficients are close to zero and statistically insignificant before VCA implementation, consistent with the parallel trends assumption. Second, the effect on the tax-inclusive price is initially small and non-significant in the first week after the reform, but increases and becomes statistically significant after two weeks—implying that the adjustment to pre-tax listed prices is gradual. Consistent with the model's predictions on dynamic adjustment, low-friction hosts (top 50% price correlation) begin adjusting their pre-tax listed prices in the first week, whereas high-friction hosts (bottom 50% correlation) adjust more gradually.

4.2 The Effect of VCA on Reservation Status

The preceding results establish that VCA adoption leads to higher tax-inclusive prices for consumers—particularly for listings operated by high-friction hosts who absorb less of the tax. A natural question is whether these tax-inclusive price increases are reflected in booking activity.

To address this, we examine the impact of VCA reform on reservation status using the following DiD specification:

$$R_{ijt} = \beta \cdot VCA_{jt} + \lambda_i + X_{jt} + \varepsilon_{ijt} \tag{3}$$

The key difference from equation (1) is the outcome variable. Here, R_{ijt} is the weekly reservation ratio in levels, calculated as the number of days booked divided by the number of days listed as available on the Airbnb platform. Following Chen and Roth (2024), we use the outcome in levels rather than $\log(1 + R_{ijt})$ because it is bounded in $[0, 1]$. The coefficient β on VCA_{jt} captures the effect of VCA reform on booking activity.

Table 3 presents the results for the weekly reservation ratio in levels. Following the structure of Table 2, Column (4) is our preferred specification using the Callaway and Sant'Anna (2021) estimator with city-specific demand controls. Panel A shows the effect on all listings, while Panels B and C detail the effects for listings with bottom and top 50% price correlation, respectively. The estimated effects on the reservation ratio are uniformly negative across panels, directionally consistent with the theoretical prediction that higher tax-inclusive prices should depress bookings. In the preferred specification (Column (4)), the pooled estimate is -0.051, the high-friction estimate is -0.074, and the low-friction estimate is -0.030. These magnitudes are larger in absolute value for high-friction listings—about 2.5 times the low-friction estimate—consistent with the pattern in Table 2, where high-friction hosts pass more of the tax through to consumers. However, the 95% wild cluster bootstrap confidence intervals all include zero in the preferred specification, so we cannot reject the null of no effect for any of the three groups; we report the point estimates here as suggestive rather than definitive evidence. The CSDID estimates without demand controls (Column (3)) are more precisely estimated—with confidence intervals that exclude zero for all three panels—and follow the same ordering, with a pooled effect of -0.138, a high-friction effect of -0.160, and a low-friction effect of -0.121.[4]

To fully investigate the heterogeneous effects of VCA reform across listings with different levels of pricing friction, we estimate equation 3 for listings categorized by decile of the local hotel price correlation distribution during the pre-VCA period. Figure 7 shows the estimated

[4]In the Appendix, we further validate that the number of listings is not changed by the VCA reform, supporting the interpretation that the change in quantity traded is driven by price adjustment rather than changes in the number of listings available.

β for each decile, along with the respective 95% confidence intervals. The bottom two deciles—the most friction-constrained hosts—show reservation-ratio declines on the order of 0.12 that are statistically distinguishable from zero. Point estimates in the middle of the distribution (deciles 3 through 8) remain negative but are estimated imprecisely, with confidence intervals that include zero; the top two deciles are centered near zero. Taken together, the decile pattern is directionally consistent with the prediction that high-friction hosts—whose consumers face the largest tax-inclusive price increases—experience the largest declines in bookings, while low-friction hosts, who absorb more of the tax, experience little or no decline. We interpret this as suggestive rather than definitive evidence for a clear welfare consequence of heterogeneous pricing frictions: hosts with greater friction plausibly shift more of the tax burden onto consumers and, in turn, appear to experience larger quantity declines than they would in a frictionless benchmark.

Similarly, to examine the common trend assumption and trace the full dynamic trajectory of the VCA's effect on reservations, we estimate the dynamic difference-in-differences specification:

$$R_{ijt} = \sum_{k=-5}^{k=12} \beta_k \cdot VCA_j \times \mathbf{1}[t = c + k] + X_{jt}\psi + \epsilon_{ijt}. \tag{4}$$

Figure 8 presents the estimated β_k coefficients from equation (4). As in the price event study, black circles denote the pooled estimate, blue triangles denote high-friction listings, and orange diamonds denote low-friction listings, with 95% confidence intervals.

Two key insights emerge. First, there is no evidence of pre-existing trends: the coefficients are close to zero and statistically insignificant before VCA implementation, consistent with the parallel trends assumption. Second, the decline in reservation intensity is gradual over the first four weeks following the reform, after which it stabilizes.

5 Theoretical Model

In this section, we model the effect of a VCA on Airbnb market prices, tax incidence,and welfare where hosts face different levels of pricing frictions.

We consider a market for differentiated properties populated by a continuum of hosts. Hosts are heterogeneous in their price-setting behavior and are categorized into three types. The introduction of a government policy is modeled as a new ad-valorem tax, τ, on host prices. We show that the presence of rigid-price firms acts as an 'anchor,' shifting the long-run tax burden from producers to consumers. This mechanism leads to a counterintuitive welfare ranking where both flexible and rigid firms in a market with frictions can earn higher profits than firms in a frictionless benchmark.

5.1 Setting

The proportion of each host type is given by α_i, where:

- **Type 1: Rigid hosts** (proportion α_1): These hosts maintain a fixed price, p_0, which was the equilibrium price before the introduction of the tax.
- **Type 2: Partially Flexible hosts** (proportion α_2): These hosts follow a Calvo-style (Calvo, 1983) pricing rule. In any given period, they can re-optimize their price with probability $\mu_2 \in (0, 1)$. With probability $1 - \mu_2$, they are constrained to charge the price they set at their last adjustment.
- **Type 3: Fully Flexible hosts** (proportion α_3): These hosts can and do re-optimize their price in every period.

The proportions sum to unity: $\sum_{i=1}^{3} \alpha_i = 1$.

Regardless of their types, hosts have an identical cost function $C(q_j)$, with positive and increasing marginal costs (i.e., $C'(q_j) > 0$, $C''(q_j) \geq 0$), where q_j denotes the quantity of host j. The quantity demanded for host j, $q_j = D(p_j^c, \bar{p}^c)$, is a function of its own consumer price, p_j^c, and the average consumer price, $\bar{p}^c$. We assume demand is downward sloping in own-price ($\frac{\partial D}{\partial p_j^c} < 0$) and increasing in the average market price ($\frac{\partial D}{\partial \bar{p}^c} > 0$), as products are substitutes.

The government enacts an *ad valorem* tax, τ, so $p_j^c = p_j(1 + \tau)$, where p_j denotes the host price. The host's profit, Π, is its revenue minus its costs, which can be expressed as a general function of its own host price p_j, the average host price $\bar{p}$, and the tax rate τ.

$$\Pi_j = \Pi(p_j, \bar{p}, \tau) = p_j \cdot D(p_j(1 + \tau), \bar{p}(1 + \tau)) - C(D(p_j(1 + \tau), \bar{p}(1 + \tau))) \tag{5}$$

Our results rely on four standard assumptions about this profit function.

Assumption 1 (Concavity)**.** The profit function is strictly concave in p_j, ensuring a unique optimal price.

$$\frac{\partial^2 \Pi}{\partial p_j^2} < 0$$

Assumption 2 (Strategic Complementarity (Supermodularity))**.** A host's marginal profitability from increasing its own price is increasing in the average market price. This is the model's core assumption.

$$\frac{\partial^2 \Pi}{\partial p_j \partial \bar{p}} > 0$$

Assumption 3 (Positive Environmental Effect:)**.** A host's profit is increasing in the average market price, as higher rival prices increase its own demand.

$$\frac{\partial \Pi}{\partial \bar{p}} > 0$$

Assumption 4 (Tax Effect (Submodularity))**.** An increase in the tax rate τ decreases the marginal profitability of raising the *host* price p_j.

$$\frac{\partial^2 \Pi}{\partial p_j \partial \tau} < 0$$

Let us turn now to the flexible host's problem. We assume the market is in equilibrium where all hosts charge p_0 ($\frac{\partial \Pi}{\partial p_j}(p_0, p_0, 0) = 0$) before the tax is enacted. After the tax is introduced, rigid hosts will persistently set their host price equal to the pre-tax price, p_0. We assume flexible and semi-flexible hosts have backward-looking adaptive expectations [5] and use the previous period's average price to form their expectations.

Then, a flexible host (Type 3, or Type 2 that gets to update) chooses its price p_t to solve:

$$\max_{p_t} \Pi(p_t, \bar{p}_{t-1}, \tau)$$

The first-order condition $\frac{\partial \Pi}{\partial p_t}(p_t, \bar{p}_{t-1}, \tau) = 0$ implicitly defines the optimal price p_t^* as a general

[5] We employ backward-looking adaptive expectations for both theoretical and empirical reasons. First, this framework reflects a plausible form of bounded rationality for the producers in this specific market (i.e., individual hosts), who are more likely to use simple heuristics based on recently observed market prices than to solve a complex, forward-looking dynamic optimization problem. Second, this assumption endogenously generates the gradual price adjustment and inertia that is commonly observed in high-frequency price data following a shock, which aligns well with our empirical investigation.

best-response function:

$$p_t^* = p^*(\bar{p}_{t-1}, \tau) \tag{6}$$

Based on our assumptions on the profit function, the optimal price has two key properties: (i) it increases in the average price, i.e., $\frac{\partial p^*}{\partial \bar{p}_{t-1}} > 0$,[6] which is a direct result of strategic complementarity, (ii) it decreases in the tax rate, $\frac{\partial p^*}{\partial \tau} < 0$.

5.2 Law of Motion

We can also characterize the law of motion for the average price. The average market price at time t, $\overline{p}_t$, aggregates the prices of all hosts. Using the standard Calvo definition (where non-updating Type 2 hosts keep their prior-period price, $p_{2,t-1}$), the average price is:

$$\overline{p}_t = \alpha_1 p_0 + \alpha_3 p_t^* + \alpha_2 p_{2,t} \tag{7}$$

where $p_{2,t}$ is the average price of Type 2 hosts, which evolves as:

$$p_{2,t} = (1 - \mu_2) p_{2,t-1} + \mu_2 p_t^* \tag{8}$$

Solving for $p_{2,t-1}$ in the $t-1$ version of the first equation and substituting it into the second yields a general, non-linear, second-order difference equation:

$$\overline{p}_t = \alpha_1 \mu_2 p_0 + (1 - \mu_2)\overline{p}_{t-1} + (\alpha_3 + \alpha_2 \mu_2) p_t^* - (1 - \mu_2)\alpha_3 p_{t-1}^* \tag{9}$$

Substituting the best-response function $p_t^* = p^*(\overline{p}_{t-1}, \tau)$ gives the full law of motion:

$$\overline{p}_t = \alpha_1 \mu_2 p_0 + (1 - \mu_2)\overline{p}_{t-1} + (\alpha_3 + \alpha_2 \mu_2) p^*(\overline{p}_{t-1}, \tau) - (1 - \mu_2)\alpha_3 p^*(\overline{p}_{t-2}, \tau) \tag{10}$$

This equation shows that the current average price $\overline{p}_t$ depends on the last two periods ($\overline{p}_{t-1}$ and $\overline{p}_{t-2}$), giving the model its dynamic persistence, or "memory."

We can also characterize the long-run equilibrium. As $t \to \infty$, the market settles into a steady state where $\bar{p}_t = \bar{p}_{t-1} = \bar{p}_\infty$ and $p_t^* = p_\infty^*$. In this equilibrium, all partially flexible hosts (Type 2) have had a chance to update, so their prices converge to the same optimal price p_∞^*

[6] We assume $\frac{\partial p^*}{\partial \bar{p}_{t-1}} < 1$ for dynamic stability.

as the flexible hosts (Type 3). The long-run price is therefore a weighted average of only the rigid hosts and all non-rigid hosts $(1-\alpha_1)$: The dynamic adjustment path (which is non-linear in this general model) converges to this long-run equilibrium.

This equilibrium is defined by the solution to the following two-equation system:

$$p^*_\infty = p^*(\bar{p}_\infty, \tau) \tag{11}$$

$$\bar{p}_\infty = \alpha_1 p_0 + (1-\alpha_1)p^*_\infty \tag{12}$$

5.3 Model Results

In this section, we present the results on welfare and tax burden. We also compare the friction model (where $\alpha_1 > 0$) to the frictionless benchmark (where $\alpha_1 = 0$).

In the frictionless world, in the absence of a tax ($\tau = 0$), a host solves $\max_{p_j} \Pi(p_j, \bar{p}, 0)$. In a symmetric equilibrium, $p_j = \bar{p} = p_0$. This pre-tax equilibrium price p_0 is the unique solution to the first-order condition:

$$\frac{\partial \Pi}{\partial p_j}(p_0, p_0, 0) = 0 \tag{13}$$

This p_0 is the price that "rigid" firms are constrained to use post-tax in the friction world.

When an ad-valorem tax is introduced, all hosts re-optimize. A firm solves $\max_{p_j} \Pi(p_j, \bar{p}, \tau)$. In the new symmetric equilibrium, $p_j = \bar{p} = p^*$. This frictionless post-tax price p^* is the solution to:

$$\frac{\partial \Pi}{\partial p_j}(p^*, p^*, \tau) = 0 \tag{14}$$

From our Tax Effect assumption (Assumption 4), it follows that the new optimal producer price is lower than the pre-tax price: $p^* < p_0$.

5.3.1 Tax Burden

Proposition 1 (Tax Burden Shift)**.** As the proportion of rigid firms (α_1) increases, the Producer Tax Burden (PTB) decreases and the Consumer Tax Burden (CTB) increases.

As α_1 increases, the higher proportion of firms locked into the pre-tax price p_0 prevents the average market price from falling as much as in the frictionless case, thereby shifting more of the tax burden onto consumers.

Corollary 1 (Producer Tax Burden Comparison)**.** For any proportion of rigid firms $\alpha_1 > 0$, the long-run Producer Tax Burden in the friction model ($PTB_{friction}$) is strictly lower than the Producer Tax Burden in the standard frictionless model ($PTB_{standard}$).

The presence of frictions that anchor prices upward reduces the relative burden on producers, as their constrained competitors partially absorb the tax impact.

5.3.2 Profits

Proposition 2 (Welfare Ranking of Firms)**.** In the post-tax, long-run equilibrium, flexible firms earn strictly higher profits and suffer a strictly smaller welfare loss than rigid firms.

Flexible firms can optimize to their market environment, while rigid firms are locked into a suboptimal price, making flexibility a valuable asset.

Proposition 3 (Benefit of Frictions for Flexible Firms)**.** A flexible firm in the friction model (with $\alpha_1 > 0$) earns a strictly higher profit than a firm in the frictionless benchmark model.

Frictions that raise the average market price provide an "environmental" benefit to all firms, allowing flexible firms to earn strictly higher profits despite the tax.

Proposition 4 (Welfare of Rigid Firms)**.** For a sufficiently small tax ($\tau > 0$), a rigid firm in the friction model earns a higher profit than a firm in the frictionless benchmark ($\Pi_{rigid} > \Pi^*$) as long as there is any positive proportion of rigid firms in the market ($\alpha_1 > 0$).

This outcome is driven by two competing effects: (1) a negative "suboptimal pricing effect" from being constrained to p_0, and (2) a positive "environmental effect" from other rigid firms raising the average price $\bar{p}_\infty$. For a small tax, the price deviation is small, making the loss a second-order effect. The environmental gain, however, is a first-order effect. The positive first-order effect dominates.

Corollary 2 (The Complete Post-Tax Welfare Ranking)**.** For a sufficiently small tax ($\tau > 0$), there is a strict welfare (profit) ordering:

$$\Pi_{flex} > \Pi_{rigid} > \Pi^*$$

This complete ranking demonstrates that frictions generate a strict hierarchy of firm profitability, with the flexibility to adjust being the primary determinant of post-tax welfare.

5.3.3 Excess Burden

For the following proposition, we need to be able to compare equilibrium quantities. To do so, we add a standard assumption: Aggregate market quantity $Q_{agg}(p^c)$ is a strictly decreasing function of the symmetric consumer price p^c.

Proposition 5 (The Inefficiency of Price Frictions)**.** For any $\tau > 0$ and $\alpha_1 > 0$, the Deadweight Loss in the friction model ($DWL_{friction}$) is strictly greater than the Deadweight Loss in the standard frictionless model ($DWL_{standard}$).

Price frictions create an additional layer of deadweight loss by maintaining a higher average price and lower equilibrium quantity compared to the frictionless benchmark.

Note on Organization: The main propositions above provide the key theoretical results concerning tax burden, firm profitability, and efficiency. For a complete treatment including comparative statics and dynamic results, as well as the full proofs of these results and supporting lemmas, please see Appendix C.

6 Results Discussion & Policy Implications

Our findings have implications for several ongoing policy debates concerning platform regulation, tax administration, and the design of compliance mechanisms in digital markets.

Efficiency costs and distributional consequences. VCAs are often presented as a policy success: they dramatically improve compliance rates and reduce administrative costs relative to individual host enforcement. And yet, there are trade-offs.

Imposing tax on markets with non-responsive suppliers is significantly less efficient (i.e. the dead weight loss is higher) compared with traditional markets. This is intuitive: if a substantial portion of the suppliers list at a sub optimally high price, there are many transactions that simply will not occur, shrinking the market size. This runs counter to the literature on unresponsive consumers: for example, when taxes are not salient, consumers will underreact to them (Chetty et al. (2009). As a result, the deadweight loss of imposing a sales tax is inversely proportional to how salient that sales tax is. Yet, in the context of Airbnb, if a host underreacts to a change in remittance obligation that functionally acts as an effective tax increase, she may end up passing through 100 percent of the tax to consumers. This, in turn, can have a large effect on consumer behavior and result in a greater deadweight loss than if the host was aware of, and responsive to, the tax.

There are also distributional considerations. Our results indicate that switching the obligation to remit to Airbnb significantly burdened consumers, who, after the policy, paid roughly 5.4 percent more for a virtually identical product—about half of the statutory tax rate change. In addition, the incidence of platform-collected taxes is not uniform across hosts. Less sophisticated hosts—those whose pricing behavior shows little responsiveness to market conditions—absorb a smaller share of the tax and instead pass the burden to their consumers through higher tax-inclusive prices. These same hosts also appear to experience larger declines in bookings in absolute magnitude, although the reservation-ratio estimates are less precisely estimated than the price effects and are statistically distinguishable from zero only in the most friction-constrained deciles. To the extent that less sophisticated hosts are also smaller, less experienced, or more casual participants in the market, VCA-style enforcement may have regressive effects within the host population, disproportionately harming those least equipped to adjust. This suggests that policymakers considering the design of tax collection mechanisms should attend to the behavioral capacity of the agents affected, not only the administrative convenience of collection.

The promise and the peril of government reliance on VCAs. Voluntary compliance agreements (VCAs) are attractive tax collection tools for local governments for two reasons. First, in the U.S., most sales taxes are imposed by state and local governments that have limited power to compel "remote" or platform sellers to remit taxes; absent a federal solution, VCAs allow these governments to recoup some of this otherwise foregone tax revenue. Second, VCAs offer an alternative to information reporting for capacity-constrained states that may be unable to collect a tax even with identifying information about the seller. However, the long-term effects of VCAs remain unclear, and may be potentially troubling. Platforms that negotiate VCAs with local governments often do so from a position of considerable market strength, and as a result they can secure significant concessions. For example, in exchange for remitting hotel taxes as a lump sum, Airbnb's VCAs with local governments do not require it to provide identifying information about hosts to the tax authorities. Not only does this prevent local tax authorities from recouping taxes owed on previous transactions from hosts directly, it also prevents them from monitoring their behavior on other platforms, including direct competitors to Airbnb, that have not signed VCAs. Put differently, VCAs can make local governments permanently dependent on the individual firm for significant revenues, and are signed when those firms have accrued sufficient market power to negotiate them on their terms.

Platform design and host pricing capacity. A striking institutional feature of this setting is that, during our study period, Airbnb deliberately refrained from providing hosts with pricing guidance—including guidance on how to adjust for newly imposed taxes—out of concern that doing so might jeopardize hosts' legal classification as independent contractors. This creates a tension: the platform centralizes tax compliance through VCAs but leaves hosts to navigate the pricing consequences of that compliance on their own. Our evidence that many hosts fail to adjust their listed prices optimally suggests that platform design choices—such as whether to offer tax-aware pricing tools, default price adjustments, or informational nudges—could meaningfully alter tax incidence. Whether platforms should be expected or required to provide such tools is an open question with implications for the broader debate over platform responsibility in the gig economy.

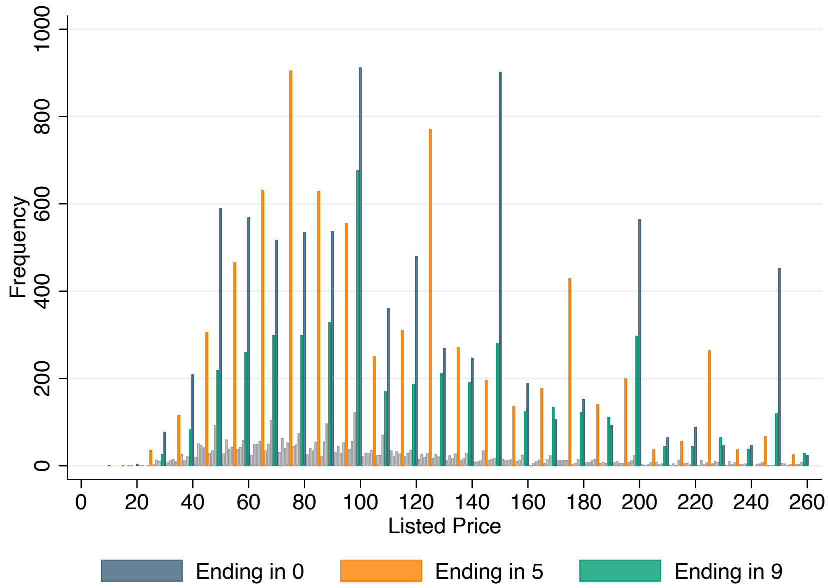
Broader applicability. While our empirical setting is specific to Airbnb and occupancy taxes, the underlying mechanism—that heterogeneous pricing frictions among suppliers alter tax incidence—is likely relevant to any market characterized by a mix of sophisticated and unsophisticated sellers. In particular, the "anchoring" effect identified in our theoretical model, whereby rigid price-setters help sustain higher market prices and shift incidence toward consumers, operates whenever firms differ in their ability or willingness to adjust prices in response to cost shocks. This includes other platform-mediated markets (e.g., ride-sharing, freelance labor, e-commerce marketplaces) as well as traditional markets with small or informal sellers, such as farmers' markets, family-owned restaurants, or owner-operated retail. As governments increasingly turn to platforms as intermediaries for tax collection and regulatory compliance, understanding how supplier frictions shape the downstream effects of these policies becomes essential.

7 Conclusion

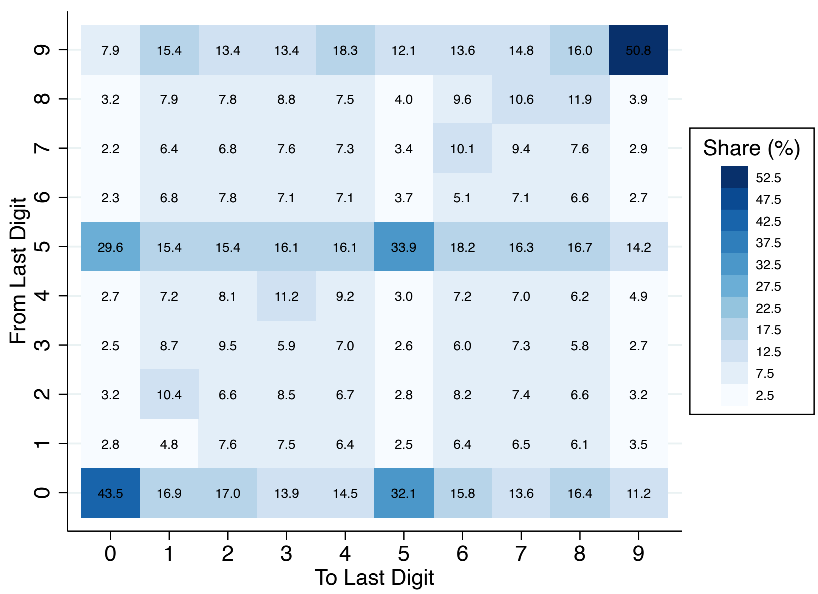
While it is widely accepted that consumer optimization frictions affect tax incidence, far less is known about whether and how supplier frictions change relative tax burdens. We contribute to closing this gap by studying producer frictions in a market where they are likely to be present: short-term listings on a large platform firm. We find that Airbnb host pricing behaviors exhibit significant frictions and that these frictions are not uniformly distributed but correlate with other measures of professionalism.

Using a natural experiment that varied the effective tax rate, we find that those hosts with a low correlation with local hotel prices before the policy passed on more of the tax burden to consumers. We build a simple model to understand the welfare consequences of our empirical findings. While we find that deadweight loss is exacerbated by the presence of supplier frictions, producer surplus for hosts facing all levels of friction is higher compared with a frictionless world.

Figure 1: Airbnb Host Pricing Heuristics



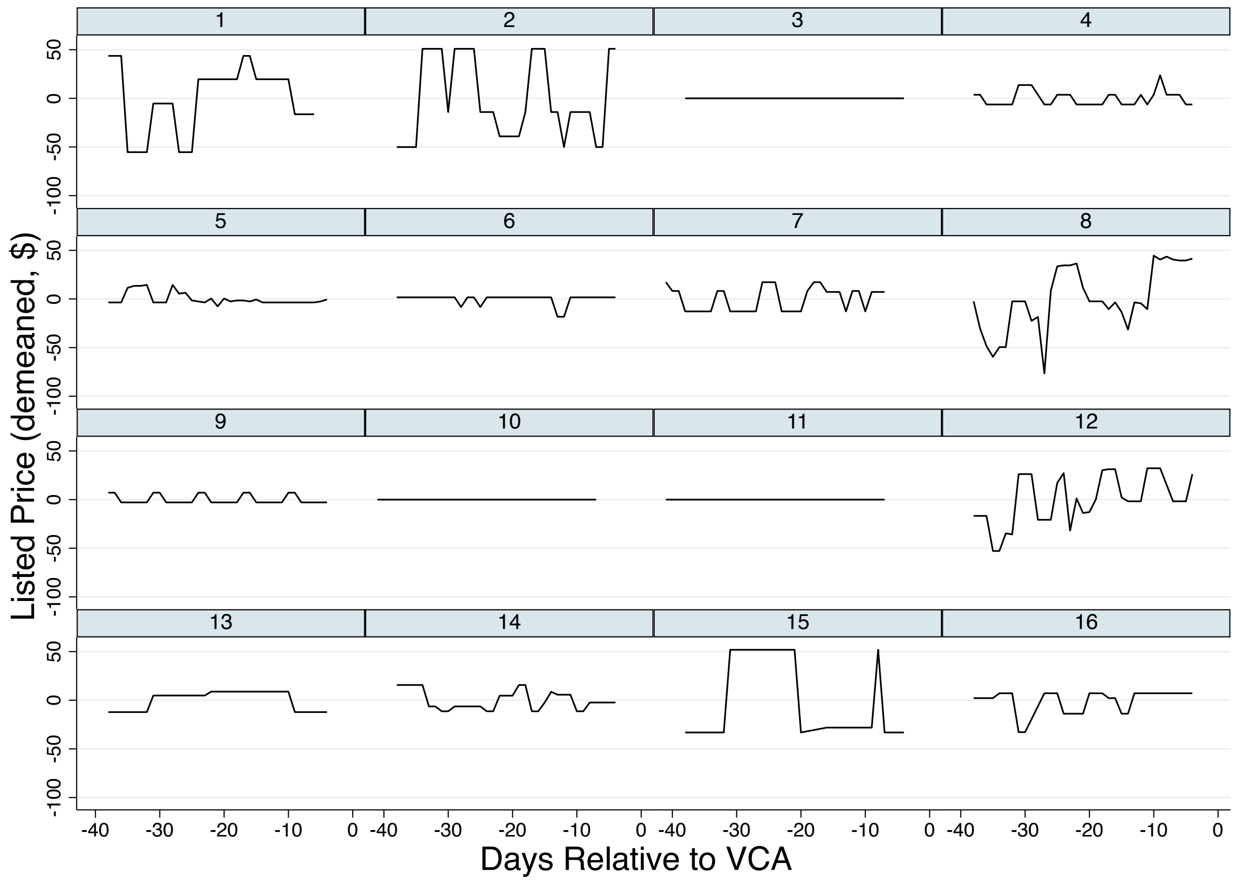
(a) Listing Prices Prior to VCA Adoption



(b) Right-most Digit Transition Matrix

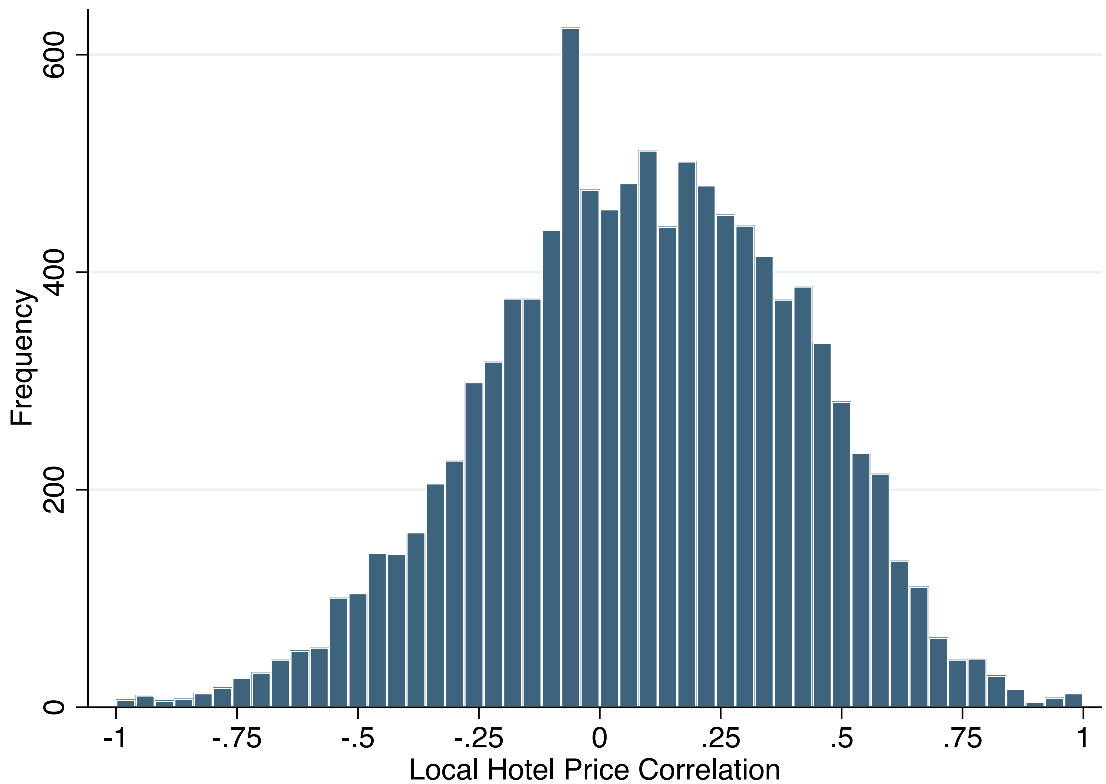
Notes: Data include all listed prices prior to VCA adoption. Panel A plots the distribution of nightly listed prices; blue bars represent prices ending in 0, orange bars prices ending in 5, and green bars prices ending in 9. Prices ending in any other digit are shown in light gray. Panel B reports a transition matrix for the right-most digit of the listed price: each cell gives the share of price changes (in %) from a given origin digit (row) to a given destination digit (column). The sample for Panel B is restricted to active price revisions in the four weeks prior to VCA adoption, dropping consecutive observations at the same listed price. Rows sum to 100%.

Figure 2: Airbnb Host Pricing Over Time



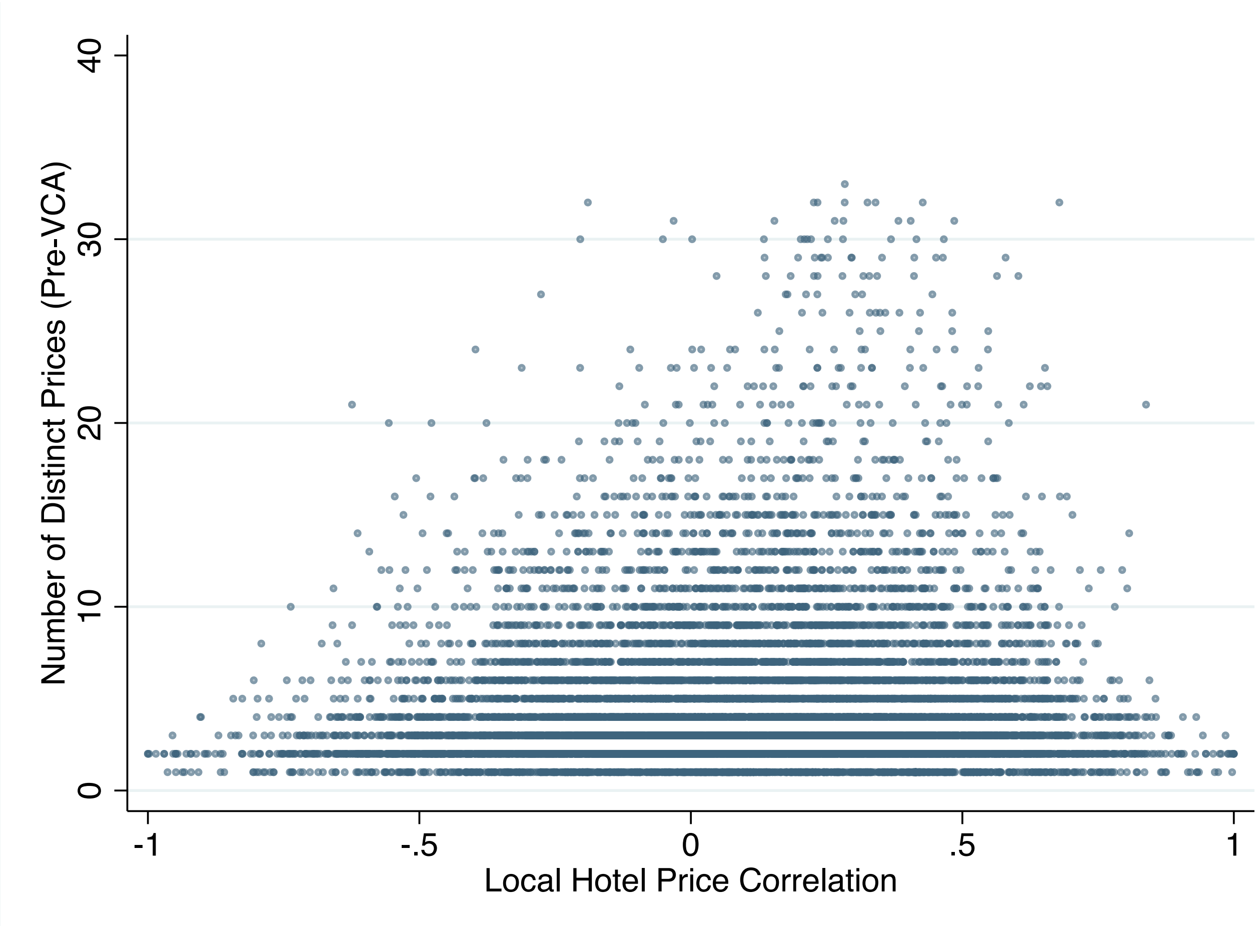
Notes: Each panel displays the daily listed price trajectory of a randomly selected Airbnb property over the ten weeks prior to VCA adoption, with prices demeaned at the property level and the horizontal axis measured in days relative to the VCA implementation date (using Phoenix's implementation date as the reference point for control properties). Properties are sampled across terciles of the pre-policy correlation between the property's listed price and local hotel prices—a measure of price-setting friction that we introduce in Figure 3 below—with 5 properties drawn from the bottom tercile, 6 from the middle, and 5 from the top, restricted to listings active on at least 30 days during the pre-period. The panels illustrate the range of pricing behavior among Airbnb hosts, from constant prices (no dynamic adjustment) to simplified weekday/weekend strategies to frequent price changes resembling hotel-style dynamic pricing.

Figure 3: Distribution of Correlations between Airbnb Properties' Listed Prices and Local Hotel Prices



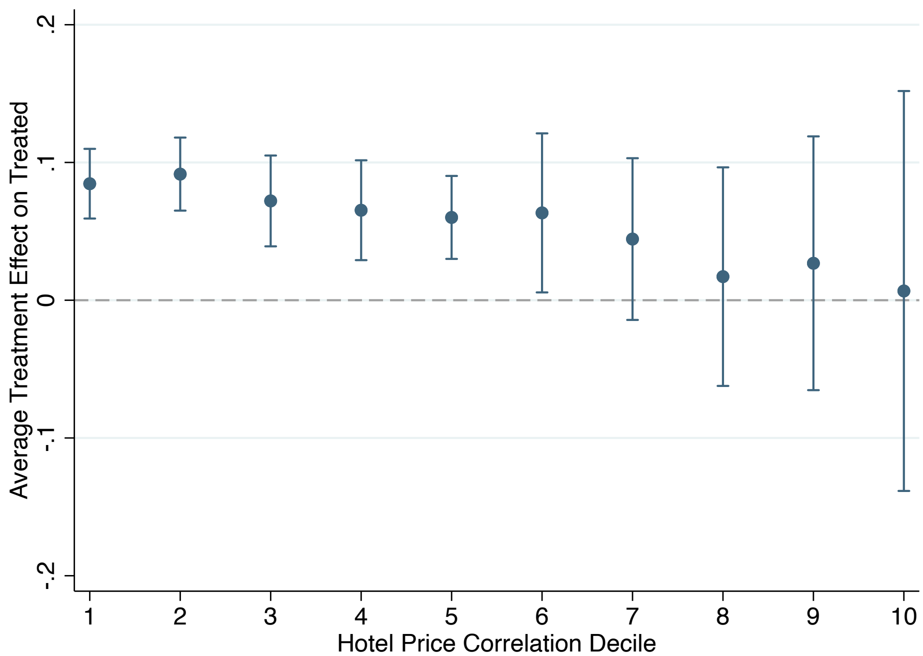
Notes: This histogram plots the distribution of the pre-policy correlation between an Airbnb property's daily listed price and the average daily price of hotels in the same MSA for properties located in treated cities, using one observation per property. The sample is restricted to the pre-policy period and excludes properties whose listed price never varied (for which the correlation is undefined). Positive correlation values indicate that the Airbnb listing price moved with local hotel pricing, i.e., that the host was responsive to local demand conditions; the distribution is centered slightly above zero, with substantial mass on both sides.

Figure 4: Validating the Hotel Price Correlation as a Measure of Pricing Friction



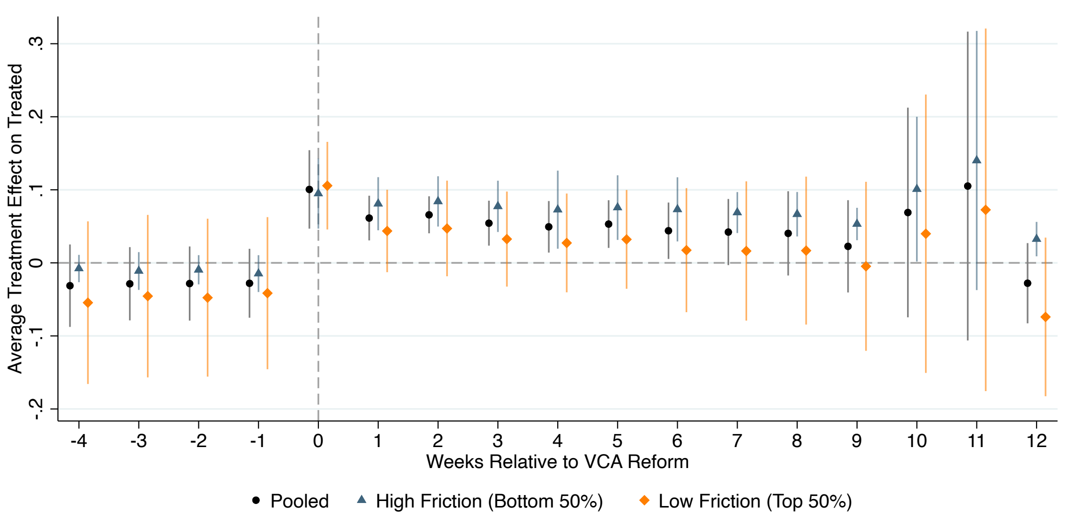
Notes: This scatter plot displays each Airbnb property in our analysis sample as a single point, with the pre-policy correlation between the property's listed price and the average price of hotels in the same MSA on the horizontal axis and the number of distinct listed prices the host posted during the pre-VCA period on the vertical axis. The sample includes both treated properties (over the pre-VCA period) and properties in control cities, with one observation per property. Properties whose listed price never varied (and for which the correlation is undefined) are excluded. Hosts with very few distinct prices mechanically cannot attain large positive or large negative correlations with hotel prices, while only hosts who actively repriced can achieve correlations near ±1, producing the triangular envelope visible in the figure.

Figure 5: The Effect of VCA on the Log Tax-Inclusive Price, by Hotel Price Correlation Decile



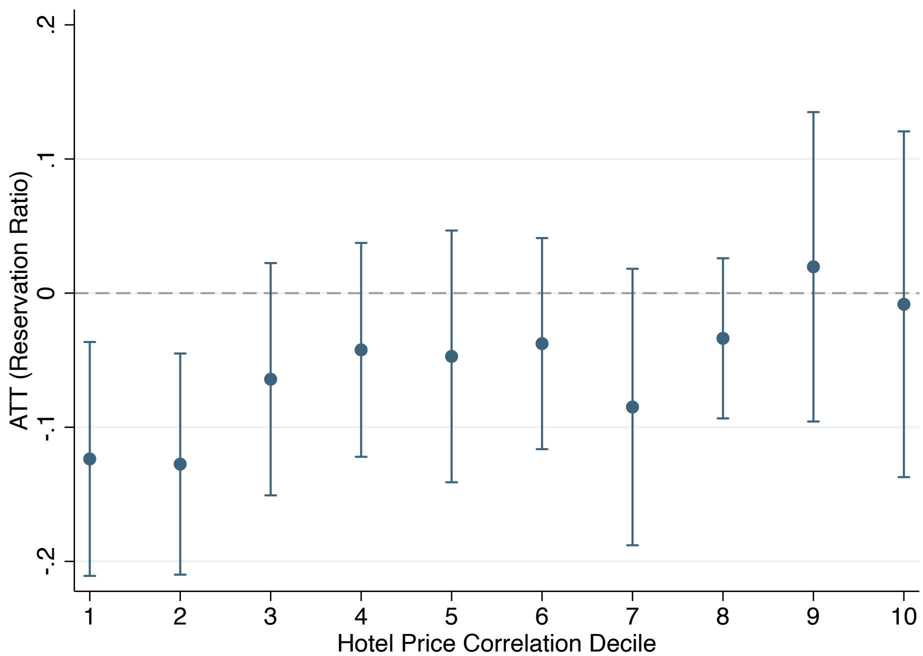
Notes: This figure plots the average treatment effect on the treated (ATT) of VCA adoption, estimated separately for each decile of the pre-policy correlation between an Airbnb property's listed price and local hotel prices. Decile 1 contains properties in the bottom 10% of the correlation distribution (the most friction-constrained hosts) and decile 10 contains those in the top 10% (the least friction-constrained). The dependent variable is the log tax-inclusive price—the total price paid by the consumer, including the listed price, cleaning fee, and service fee, plus any applicable occupancy taxes. Each ATT is estimated using the Callaway and Sant'Anna (2021) estimator with the doubly-robust IPW estimator, never-treated cities as the comparison group, a universal base period, city-specific year-week demand controls, property fixed effects, and a wild cluster bootstrap clustered at the MSA level. The bars indicate 95% confidence intervals.

Figure 6: Event Study: Effect of VCA on the Log Tax-Inclusive Price



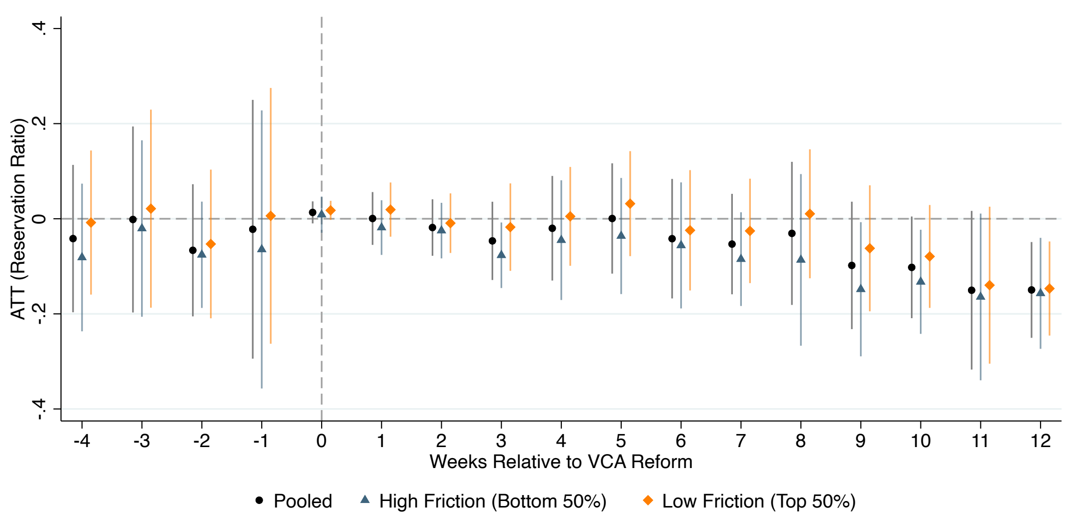
This figure reports dynamic average treatment effects on the treated (ATT) at each event time relative to VCA adoption, estimated using the Callaway and Sant'Anna (2021) doubly-robust IPW estimator with the never-treated as the comparison group and a universal base period. The dependent variable is the log tax-inclusive price, defined as the total price paid by the consumer including the listed price, cleaning fee, and service fee, as well as any applicable occupancy taxes. The pooled series (black circles) uses all properties in treated cities; the high-friction series (blue triangles) restricts to properties whose pre-VCA price series has a hotel-price correlation below the MSA-specific median, and the low-friction series (orange diamonds) restricts to those above the median. All specifications include the demand control. Series are jittered horizontally to avoid overlap. The bars indicate 95% confidence intervals based on analytical standard errors clustered at the MSA level. (The pooled simple ATTs reported in Table **??** and the percentile estimates in Figure **??** use a wild cluster bootstrap; we use analytical clustered SEs here because per-event-time wild bootstrap CIs were unstable with $G = 10$ clusters.)

Figure 7: The Effect of VCA on the Reservation Ratio, by Hotel Price Correlation Decile



This figure plots the pooled average treatment effect on the treated (ATT) for the reservation ratio, separately for each decile of the local hotel price correlation distribution. Decile 1 corresponds to properties with the lowest pre-VCA correlation between their listed price and the local hotel price index (the most "high friction" pricers), and decile 10 to the highest correlation (the most "low friction" pricers). ATTs are estimated using the Callaway and Sant'Anna (2021) doubly-robust IPW estimator with the never-treated as the comparison group and a universal base period, including the demand control. The dependent variable is the reservation ratio in levels, defined as the number of days booked divided by the number of days listed as available; following Chen and Roth (2024), we use levels rather than $\log(1 + \text{ratio})$ because the outcome is bounded in $[0, 1]$. The bars indicate 95% confidence intervals based on a wild cluster bootstrap clustered at the MSA level.

FIGURE 8: Event Study: Effect of VCA on the Reservation Ratio



This figure reports dynamic average treatment effects on the treated (ATT) at each event time relative to VCA adoption, estimated using the Callaway and Sant'Anna (2021) doubly-robust IPW estimator with the never-treated as the comparison group and a universal base period. The dependent variable is the reservation ratio in levels, defined as the number of days booked divided by the number of days listed as available; following Chen and Roth (2024), we use levels rather than $\log(1 + \text{ratio})$ because the outcome is bounded in $[0, 1]$. The pooled series (black circles) uses all properties in treated cities; the high-friction series (blue triangles) restricts to properties whose pre-VCA price series has a hotel-price correlation below the MSA-specific median, and the low-friction series (orange diamonds) restricts to those above the median. All specifications include the demand control. Series are jittered horizontally to avoid overlap. The bars indicate 95% confidence intervals based on analytical standard errors clustered at the MSA level. (The pooled simple ATTs reported in Table **??** and the percentile estimates in Figure 7 use a wild cluster bootstrap; we use analytical clustered SEs here because per-event-time wild bootstrap CIs were unstable with $G = 10$ clusters.)

Table 1: Descriptive Statistics

	All Properties			High Friction			Low Friction		
	Mean	SD	Median	Mean	SD	Median	Mean	SD	Median
Panel A: Treated Cities									
Multi-Property Host (%)	0.46	0.49	0.00	0.51	0.49	0.80	0.40	0.48	0.00
Entire Home (%)	0.63	0.48	1.00	0.67	0.47	1.00	0.59	0.49	1.00
Bathrooms	1.22	0.47	1.00	1.22	0.46	1.00	1.21	0.48	1.00
Bedrooms	1.14	0.72	1.00	1.16	0.73	1.00	1.13	0.70	1.00
Listed Price ($)	126.21	70.18	106.83	131.04	70.25	113.17	120.11	69.63	100.00
Tax-Inclusive Price ($)	172.62	97.04	150.00	181.33	98.23	161.32	161.62	94.38	136.72
Reservation Ratio	0.63	0.32	0.74	0.66	0.30	0.77	0.59	0.33	0.66
Average Tax Rate Change	0.12	0.03	0.14	0.12	0.03	0.14	0.12	0.03	0.14
Properties		5,658			3,159			2,499	
	Mean	SD	Median	Mean	SD	Median	Mean	SD	Median
Panel B: Control Cities									
Multi-Property Host (%)	0.22	0.41	0.00	0.16	0.36	0.00	0.26	0.43	0.00
Entire Home (%)	0.68	0.47	1.00	0.62	0.48	1.00	0.72	0.45	1.00
Bathrooms	1.33	0.58	1.00	1.33	0.58	1.00	1.33	0.58	1.00
Bedrooms	1.49	0.89	1.00	1.46	0.87	1.00	1.51	0.90	1.00
Listed Price ($)	166.41	165.23	116.43	183.25	165.67	125.00	155.23	164.03	113.97
Tax-Inclusive Price ($)	189.63	150.91	145.00	176.88	156.75	125.00	193.68	148.82	150.00
Reservation Ratio	0.25	0.32	0.09	0.13	0.26	0.00	0.33	0.33	0.24
Properties		3,465			1,382			2,083	

Notes: This table presents descriptive statistics for treated cities (Boulder, Philadelphia, Phoenix, San Diego, and Los Angeles) and control cities (Austin, Dallas, Houston, New Orleans, and Savannah). Properties are classified by pricing friction based on the pre-policy correlation between the property's listed price and local hotel prices. "High Friction" refers to properties whose correlation falls below the sample median; "Low Friction" refers to those at or above the median. The reservation ratio is defined as the number of days booked divided by the number of days listed as available. Listed Price is the nightly price set by the host, exclusive of taxes; Tax-Inclusive Price is the total price paid by the consumer.

Table 2: Staggered DID Estimation: Log Tax-Inclusive Price

	(1) TWFE (no demand)	(2) TWFE (w/ demand)	(3) CSDID (no demand)	(4) CSDID (w/ demand)
Panel A: All listings				
VCA × Post	0.092*	0.100*	0.079*	0.054*
	[0.059, 0.129]	[0.076, 0.131]	[0.051, 0.107]	[0.006, 0.101]
Observations	346,301	346,301	325,833	188,085
Panel B: High friction (bottom 50% correlation)				
VCA × Post	0.102*	0.107*	0.116*	0.079*
	[0.058, 0.139]	[0.070, 0.140]	[0.090, 0.143]	[0.045, 0.113]
Observations	154,825	154,825	143,012	79,666
Panel C: Low friction (top 50% correlation)				
VCA × Post	0.083*	0.094*	0.049*	0.031
	[0.049, 0.123]	[0.067, 0.128]	[0.010, 0.088]	[-0.053, 0.114]
Observations	191,476	191,476	182,821	108,419

Notes: This table presents DiD estimates of the effect of VCA adoption on the log tax-inclusive price—the total nightly price paid by the consumer, equal to the host's listed price plus any applicable occupancy taxes collected and remitted under the VCA. Column (1) is a two-way fixed-effects model with property and year-week fixed effects; column (2) adds city-specific weekly demand controls. Columns (3) and (4) use the Callaway and Sant'Anna (2021) doubly-robust IPW estimator with the never-treated as the comparison group and a universal base period; column (4) adds the demand control. Panel A reports results for all properties; Panel B restricts to properties whose pre-VCA hotel price correlation is below the MSA-specific median (high friction); Panel C restricts to those above (low friction). 95% confidence intervals from a wild cluster bootstrap clustered at the MSA level are reported in brackets. For the TWFE columns, the bootstrap is implemented via `boottest` with Webb weights and 9,999 replications; for the CSDID columns, it is implemented via `csdid_stats simple, wboot`. * indicates that the 95% wild cluster bootstrap CI excludes zero.

Table 3: Staggered DID Estimation: Reservation Ratio

	(1) TWFE (no demand)	(2) TWFE (w/ demand)	(3) CSDID (no demand)	(4) CSDID (w/ demand)
Panel A: All listings				
VCA × Post	-0.087	-0.058	-0.138*	-0.051
	[-0.151, 0.006]	[-0.116, 0.012]	[-0.213, -0.062]	[-0.143, 0.042]
Observations	346,301	346,301	325,833	188,085
Panel B: High friction (bottom 50% correlation)				
VCA × Post	-0.090	-0.061	-0.160*	-0.074
	[-0.154, 0.002]	[-0.121, 0.007]	[-0.230, -0.089]	[-0.165, 0.016]
Observations	154,825	154,825	143,012	79,666
Panel C: Low friction (top 50% correlation)				
VCA × Post	-0.085	-0.056	-0.121*	-0.030
	[-0.150, 0.011]	[-0.113, 0.017]	[-0.200, -0.041]	[-0.122, 0.062]
Observations	191,476	191,476	182,821	108,419

Notes: This table presents DiD estimates of the effect of VCA adoption on the reservation ratio in levels, defined as the number of days booked divided by the number of days listed as available per week. Following Chen and Roth (2024), we use the outcome in levels rather than $\log(1 + \text{ratio})$ because the outcome is bounded in $[0, 1]$. Column (1) is a two-way fixed-effects model with property and year-week fixed effects; column (2) adds city-specific weekly demand controls. Columns (3) and (4) use the Callaway and Sant'Anna (2021) doubly-robust IPW estimator with the never-treated as the comparison group and a universal base period; column (4) adds the demand control. Panel A reports results for all properties; Panel B restricts to properties whose pre-VCA hotel price correlation is below the MSA-specific median (high friction); Panel C restricts to those above (low friction). 95% confidence intervals from a wild cluster bootstrap clustered at the MSA level are reported in brackets. For the TWFE columns, the bootstrap is implemented via `boottest` with Webb weights and 9,999 replications; for the CSDID columns, it is implemented via `csdid_stats simple, wboot`. * indicates that the 95% wild cluster bootstrap CI excludes zero.

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Appendix: For Online Publication

A Additional Tables and Figures

Table A1: Summary of Cities in the Analysis

	Properties	Observations	Avg. Listed Price ($)	Tax Rate (%)	VCA Effective Date
Treated Cities					
Boulder	868	12,990	132.90	7.5	10/1/2016
Philadelphia	1,208	17,348	110.40	8.5	7/15/2015
Phoenix	244	3,790	99.46	5.0	7/1/2015
San Diego	2,229	31,474	149.38	10.5	7/15/2015
Los Angeles	8,419	124,812	123.40	14.0	8/1/2016
Control Cities					
Austin	5,846	156,079	190.48		
Dallas	1,037	24,873	102.54		
Houston	1,768	44,976	95.78		
New Orleans	3,534	104,906	139.97		
Savannah	463	14,258	163.77		

Notes: This table summarizes the cities in our analysis sample. "Properties" is the number of unique listings; "Observations" is the total number of property-week observations. "Avg. Listed Price" is the mean pre-policy nightly listed price in dollars. "Tax Rate" is the local occupancy tax rate applied under the VCA. Control cities did not adopt VCAs during the sample period.

B Robustness of Price Friction

Table A2: Staggered DID Estimation: Log Tax-Inclusive Price (Traded)

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Without Instant-book-Feature						
VCA Reform	.0842***	.0926***	.0663***	.0890***	.0729***	.0482***
	(.0085)	(.0065)	(.0070)	(.0084)	(.0166)	(.0051)
	234656	234656	48815.	222957	218724	121368
Panel B: With Instant-book-Feature						
VCA Reform	.0690***	.0824***	0800***	.0821***	.0547***	.0372***
	(.0099)	(.0069)	(.0033)	(.0126)	(.0191)	(.0083)
	57566	57566	222957	48815	55008	22411
Panel C: Single Host						
VCA Reform	.0865***	.0940***	.0854***	.0911***	.0787***	.0422***
	(.0108)	(.0084)	(.0019)	(.0052)	(.0143)	(.0047)
	147431	147431	139052	139052	137066	64513
Panel D: Multiple Host						
VCA Reform	.0763***	.0874***	.0705***	.0865***	.0615***	.0499***
	(.0085)	(.0067)	(.0060)	(.0145)	(.0198)	(.0067)
	144791	144791	132720	132720	136666	79266
Estimator	TWFE	TWFE	Multiplegt	Multiplegt	CSDID	CSDID
Year-Week FE	√	√	√	√	√	√
Property FE	√	√	√	√	√	√
Cities-Weekly Demand		√		√		√
Cluster	City	City	City	City	City	City

Notes: This table presents DiD estimates using various specifications. Column (1) provides the basic DID estimate from a two-way fixed effect model, while columns (2) add city-specific year-week demand controls to capture seasonal changes. Columns (3) and (4) utilize staggered DID methods following Sun and Abraham (2021) and Columns (5) and (6) use an estimator following Callaway and Sant'Anna (2021). The dependent variable is the log consumer-paid price, encompassing the booked price plus cleaning and service fees. Note that booking prices are tax-inclusive, while the observed values are tax-exclusive. Panel A reports results for all properties, Panel B for properties in the bottom 50%, and Panel C for those in the top 50%. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control, and are clustered at the MSA level. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. Standard errors are clustered at the firm-level and reported in parentheses. *** significant at the 1 percent level, ** significant at the 5 percent level, and * significant at the 10 percent level.

Table A3: Staggered DID Estimation: Log Reservation Ratio

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Without Instant-book-Feature						
VCA Reform	-.072***	-.048**	-.096***	-.084***	-.110***	-.053**
	(.0222)	(.0202)	(.0097)	(.0298)	(.0365)	(.0270)
	234656	234656	48815	222957	218724	121368
Panel B: With Instant-book-Feature						
VCA Reform	-.048**	-.024	-.118***	-.052	-.086***	-.034
	(.0192)	(.0203)	(.0081)	(.0430)	(.0260)	(.0241)
	57566	57566	222957	48815	55008	22411
Panel C: Single Host						
VCA Reform	-.067***	-.042**	-.117***	-.085***	-.109***	-.033
	(.0234)	(.0217)	(.0057)	(.0270)	(.0307)	(.0270)
	147431	147431	139052	139052	137066	64513
Panel D: Multiple Host						
VCA Reform	-.067***	-.043**	-.107***	-.068*	-.100***	-.062***
	(.0193)	(.0181)	(.0101)	(.0361)	(.0314)	(.0235)
	144791	144791	132720	132720	136666	79266
Estimator	TWFE	TWFE	Multiplegt	Multiplegt	CSDID	CSDID
Year-Week FE	√	√	√	√	√	√
Property FE	√	√	√	√	√	√
Cities-Weekly Demand		√		√		√
Cluster	City	City	City	City	City	City

Notes: This table presents DiD estimates using various specifications. Column (1) provides the basic DID estimate from a two-way fixed effect model, while columns (2) add city-specific year-week demand controls to capture seasonal changes. Columns (3) and (4) utilize staggered DID methods following Sun and Abraham (2021) and Columns (5) and (6) use an estimator following Callaway and Sant'Anna (2021). The dependent variable is the log consumer-paid price, encompassing the booked price plus cleaning and service fees. Note that booking prices are tax-inclusive, while the observed values are tax-exclusive. Panel A reports results for all properties, Panel B for properties in the bottom 50%, and Panel C for those in the top 50%. All estimates include year-week fixed effect, property fixed effect, and MSA-Demand control, and are clustered at the MSA level. The baseline mean is the mean of the outcome for the property in the treated cities 5 weeks before the VCA reform. Standard errors are clustered at the firm-level and reported in parentheses. *** significant at the 1 percent level, ** significant at the 5 percent level, and * significant at the 10 percent level.

C Model Proofs and Additional Results

C.1 Proofs of Main Results

Proof of Proposition 1 (Tax Burden Shift).

Proof. As defined in the tax literature $PTB = p_0 - \bar{p}_\infty$ and $CTB = \bar{p}_\infty(1+\tau) - p_0$. The proof rests on showing that $\bar{p}_\infty$ is an increasing function of α_1. The steady state is implicitly defined by Equation 12::

$$\bar{p}_\infty = \alpha_1 p_0 + (1-\alpha_1)p^*(\bar{p}_\infty, \tau)$$

Let $F(\bar{p}_\infty, \alpha_1) = \bar{p}_\infty - \alpha_1 p_0 - (1-\alpha_1)p^*(\bar{p}_\infty, \tau) = 0$. By the Implicit Function Theorem (IFT), we know

$$\frac{d\bar{p}_\infty}{d\alpha_1} = -\frac{\partial F/\partial \alpha_1}{\partial F/\partial \bar{p}_\infty} \tag{15}$$

The partial derivatives are:

- $\frac{\partial F}{\partial \alpha_1} = p^*_\infty - p_0$
- $\frac{\partial F}{\partial \bar{p}_\infty} = 1 - (1-\alpha_1)\frac{\partial p^*}{\partial \bar{p}_\infty}$

Substituting these into Equation 15 gives:

$$\frac{d\bar{p}_\infty}{d\alpha_1} = \frac{p_0 - p^*_\infty}{1 - (1-\alpha_1)\frac{\partial p^*}{\partial \bar{p}_\infty}} \tag{16}$$

By the supermodularity assumption (Assumption 2) $0 < \frac{\partial p^*}{\partial \bar{p}_\infty} < 1$. Thus, the denominator is $1 -$ (a value < 1), which is positive. Therefore, to show that Equation 16 is positive, we must show $p_0 > p^*_\infty$. By definition, $p_0 = p^*(p_0, 0)$ and $p^*_\infty = p^*(\bar{p}_\infty, \tau)$. Since $\bar{p}_\infty \leq p_0$ and $\tau > 0$, by Assumption 2 we have,

$p^*_\infty = p^*(\bar{p}_\infty, \tau) \leq p^*(p_0, \tau)$. Then by the tax effect (Assumption 4) $p^*(p_0, \tau) < p^*(p_0, 0) = p_0$. Thus, $p_0 - p^*_\infty$ is **positive**. Therefore,Equation 16 is positive and $\bar{p}_\infty$ is strictly increasing in α_1.

Since $\frac{d\bar{p}_\infty}{d\alpha_1} > 0$, we can conclude that:

- $\frac{d(PTB)}{d\alpha_1} = -\frac{d\bar{p}_\infty}{d\alpha_1} < 0$. (Producer burden decreases)

- $\frac{d(CTB)}{d\alpha_1} = (1+\tau)\frac{d\bar{p}_\infty}{d\alpha_1} > 0$. (Consumer burden increases)

$\square$

Proof of Corollary 1 (Producer Tax Burden Comparison).

Proof. Let $PTB_{standard} = p_0 - p^*$, $PTB_{friction} = p_0 - \overline{p}_\infty$. As established in the proof of Proposition 1, the long-run average price $\overline{p}_\infty$ is a strictly increasing function of α_1, with a lower bound of p^* at $\alpha_1 = 0$. Therefore, for any $\alpha_1 > 0$:

$$\overline{p}_\infty > p^*$$

Since $\overline{p}_\infty > p^*$, it follows that $-\overline{p}_\infty < -p^*$. Adding p_0 to both sides of the inequality gives:

$$p_0 - \overline{p}_\infty < p_0 - p^*$$

Therefore, $PTB_{friction} < PTB_{standard}$. The producers in the friction model bear a strictly smaller tax burden. $\square$

Proof of Proposition 2 (Welfare Ranking of Firms).

Proof. Both firm types operate in the same market environment $(\bar{p}_\infty, \tau)$. A flexible firm chooses $p^*_\infty = \text{argmax}_{p_j} \Pi(p_j, \bar{p}_\infty, \tau)$. A rigid firm is constrained to p_0. As shown in Prop. 1, $p_0 \neq p^*_\infty$, so p_0 is suboptimal. By definition of maximization and concavity (Assumption 1):

$$\Pi_{flex} = \Pi(p^*_\infty, \bar{p}_\infty, \tau) > \Pi(p_0, \bar{p}_\infty, \tau) = \Pi_{rigid}$$

$\square$

Proof of Proposition 3 (Benefit of Frictions for Flexible Firms).

Proof. Let $\Pi_{max}(\bar{p}, \tau)$ be the flexible firm's value function. By the Envelope Theorem, the derivative of this function with respect to the average market price is:

$$\frac{d\Pi_{max}}{d\bar{p}} = \frac{\partial \Pi}{\partial \bar{p}}$$

By Assumption 3 (Positive Environmental Effect), $\frac{\partial \Pi}{\partial \bar{p}} > 0$. Thus, Π_{max} is a strictly increasing function of the average market price $\bar{p}$.

From the proof of Proposition 1, $\bar{p}_\infty(\alpha_1)$ is strictly increasing in α_1 with the lower bound being $\bar{p}_\infty(0) = p^*$. Therefore, for any $\alpha_1 > 0$, the average price $\bar{p}_\infty$ is strictly greater than the frictionless price p^*.

$$\bar{p}_\infty > p^* \quad (\text{for } \alpha_1 > 0)$$

Since Π_{max} is strictly increasing in the average price, and the average price is strictly higher in the friction model, the profit must also be strictly higher.

$$\Pi_{flex} = \Pi_{max}(\bar{p}_\infty, \tau) > \Pi_{max}(p^*, \tau) = \Pi^*$$

□

Lemma 1 (The Anchoring Effect)**.** The average producer price in the market with frictions ($\bar{p}_\infty$) falls more slowly (or less) than the producer price in the standard frictionless market (p^*).

Proof. We need to show that at $\tau = 0$, the derivative of the average producer price with respect to the tax is strictly greater (less negative) than the derivative of the frictionless price, for any $\alpha_1 > 0$.

$$\left.\frac{d\bar{p}_\infty}{d\tau}\right|_{\tau=0} > \left.\frac{dp^*}{d\tau}\right|_{\tau=0}$$

For a small tax, the long-run equilibrium optimal price p^*_∞ can be approximated by the frictionless price $p^*(\tau)$. The aggregation equation is:

$$\bar{p}_\infty(\tau) = \alpha_1 p_0 + (1-\alpha_1)p^*_\infty(\tau) \approx \alpha_1 p_0 + (1-\alpha_1)p^*(\tau)$$

Differentiate this approximation with respect to τ (noting p_0 is a constant):

$$\frac{d\bar{p}_\infty}{d\tau} \approx (1-\alpha_1)\frac{dp^*}{d\tau}$$

Substitute this into the difference equation, $G'_\tau(0) = (\frac{d\bar{p}_\infty}{d\tau} - \frac{dp^*}{d\tau})|_{\tau=0}$. :

$$G'_\tau(0) \approx \left.\left((1-\alpha_1)\frac{dp^*}{d\tau} - \frac{dp^*}{d\tau}\right)\right|_{\tau=0} = -\alpha_1 \left.\frac{dp^*}{d\tau}\right|_{\tau=0}$$

Note that by Assumption 4 (Tax Effect), $\frac{dp^*}{d\tau}|_{\tau=0} < 0$ and since $\alpha_1 > 0$, $G'_\tau(0) > 0$

□

Proof of Proposition 4 (Welfare of Rigid Firms).

Proof. Let $\Delta\Pi(\tau) = \Pi_{rigid}(\tau) - \Pi^*(\tau)$. At $\tau = 0$, $\Delta\Pi(0) = 0$. By chain rule:

$$\frac{d\Pi_{rigid}}{d\tau} = \frac{\partial \Pi}{\partial \bar{p}_\infty}\frac{d\bar{p}_\infty}{d\tau} + \frac{\partial \Pi}{\partial \tau} \tag{17}$$

By Envelope Theorem

$$\frac{d\Pi^*}{d\tau} = \frac{\partial \Pi}{\partial \bar{p}}\bigg|_{p^*}\frac{dp^*}{d\tau} + \frac{\partial \Pi}{\partial \tau}\bigg|_{p^*} \tag{18}$$

The derivative of the profit difference at $\tau = 0$ is as follows

$$\frac{d(\Delta\Pi)}{d\tau}\bigg|_{\tau=0} = \left(\frac{d\Pi_{rigid}}{d\tau} - \frac{d\Pi^*}{d\tau}\right)\bigg|_{\tau=0} \tag{19}$$

and at $\tau = 0$, $\bar{p}_\infty = p^* = p_0$, so all partial derivatives $\frac{\partial \Pi}{\partial \bar{p}}$ and $\frac{\partial \Pi}{\partial \tau}$ are identical.

$$\frac{d(\Delta\Pi)}{d\tau}\bigg|_{\tau=0} = \left(\frac{\partial \Pi}{\partial \bar{p}}\frac{d\bar{p}_\infty}{d\tau} + \frac{\partial \Pi}{\partial \tau}\right) - \left(\frac{\partial \Pi}{\partial \bar{p}}\frac{dp^*}{d\tau} + \frac{\partial \Pi}{\partial \tau}\right) \tag{20}$$

$$\frac{d(\Delta\Pi)}{d\tau}\bigg|_{\tau=0} = \frac{\partial \Pi}{\partial \bar{p}}\bigg|_{\tau=0} \cdot \left(\frac{d\bar{p}_\infty}{d\tau} - \frac{dp^*}{d\tau}\right)\bigg|_{\tau=0} \tag{21}$$

Recall that $\frac{\partial \Pi}{\partial \bar{p}} > 0$ by Assumption 3 and $\left(\frac{d\bar{p}_\infty}{d\tau} - \frac{dp^*}{d\tau}\right)\Big|_{\tau=0} > 0$ by Lemma 1. Since $\Delta\Pi(0) = 0$ and $\Delta\Pi'(0) > 0$, $\Delta\Pi(\tau)$ must be positive for a sufficiently small $\tau > 0$. □

Proof of Corollary 2 (The Complete Post-Tax Welfare Ranking).

Proof. This corollary summarizes the previous three general-model proofs.

- $\Pi_{flex} > \Pi_{rigid}$ (from Proposition 2).
- $\Pi_{rigid} > \Pi^*$ (from Proposition 4, for small τ).
- $\Pi_{flex} > \Pi^*$ (from Proposition 3).

□

Proof of Proposition 5 (The Inefficiency of Price Frictions).

Proof. We prove this by showing that the friction model has a higher tax wedge and a lower quantity.

The wedges are $\tau\bar{p}_\infty$ and τp^*. From Proposition 3, we proved that for any $\alpha_1 > 0$, $\bar{p}_\infty > p^*$. Since $\tau > 0$, the tax wedge is strictly larger: $(\tau\bar{p}_\infty) > (\tau p^*)$.

The long-run average consumer prices in both settings are as follows: $p_c^{flex} = \bar{p}_\infty(1+\tau)$ and $p_c^* = p^*(1+\tau)$. Since $\bar{p}_\infty > p^*$, it is clear $p_c^{flex} > p_c^*$. By the assumption we made earlier, Q_{agg} is a strictly decreasing function. Therefore:

$$q_{avg} = Q_{agg}(p_c^{flex}) < Q_{agg}(p_c^*) = q^*$$

The average quantity in the friction model is strictly smaller, implying a strictly larger reduction in quantity from the pre-tax level.

The friction model has both a strictly larger tax wedge and a strictly larger reduction in quantity, hence, $DWL_{friction} > DWL_{standard}$

□

C.2 Comparative Statics

Proposition 6 (Flexible Firm vs. Standard Case)**.** The profit advantage that a flexible firm holds over a firm in the standard model, $\Delta\Pi_{flex} = \Pi_{flex} - \Pi^*$, is a strictly increasing function of the tax rate τ.

Proof. As proven in Proposition 3, $\Pi_{max}(\bar{p}, \tau)$ is strictly increasing in $\bar{p}$. Define $G_p(\tau) = \bar{p}_\infty(\tau) - p^*(\tau)$. This gap is positive for $\alpha_1 > 0$, and as proven in Lemma 1, this difference is positive. The profit advantage is $\Delta\Pi_{flex} = \Pi_{max}(\bar{p}_\infty, \tau) - \Pi_{max}(p^*, \tau)$. Since Π_{max} is an increasing function and its argument's advantage, $G_p(\tau)$, is also increasing in τ, the profit advantage $\Delta\Pi_{flex}$ must be a strictly increasing function of τ.

□

Proposition 7 (Non-Monotonicity of Rigid Firm Profit)**.** The effect of the tax rate on the profit difference $\Delta\Pi_{rigid} = \Pi_{rigid} - \Pi^*$ initially increases for small taxes, but must eventually decrease for larger taxes.

Proof. We need to show that

1. The difference is initially increasing: $\frac{d(\Delta\Pi_{rigid})}{d\tau}|_{\tau=0} > 0$.
2. The difference must eventually decrease, as $\lim_{\tau\to\infty} \Delta\Pi_{rigid}(\tau) = 0$.

Since the function starts at 0, increases, and must return to 0, there must exist at least one threshold $\tilde{\tau} > 0$ such that for all $\tau > \tilde{\tau} > 0$, the function is decreasing.

1. As proven in Proposition 4, the positive "environmental" effect (a first-order gain from $\bar{p}_\infty > p^*$) dominates the "suboptimal pricing" effect (a second-order loss from p_0 being near the optimum) hence the derivative of the profit difference at $\tau = 0$ is strictly positive:

$$\left.\frac{d(\Delta\Pi_{rigid})}{d\tau}\right|_{\tau=0} = \underbrace{\left(\frac{\partial \Pi}{\partial \bar{p}}\right)}_{>0} \cdot \underbrace{\left(\frac{d\bar{p}_\infty}{d\tau} - \frac{dp^*}{d\tau}\right)}_{>0 \text{ (by Lemma 1)}} > 0$$

This proves that the function is initially increasing.

2. We now analyze the limit of $\Delta\Pi_{rigid}(\tau)$ as $\tau \to \infty$. For any given producer price $p_j > 0$, the consumer price is $p_j^c = p_j(1+\tau)$. As $\tau \to \infty$, $p_j^c \to \infty$. It is standard to assume $\lim_{p^c \to \infty} D(p_j^c, \bar{p}^c) = 0$, which in turn implies the quantity sold for any firm j converges to zero: $\lim_{\tau \to \infty} q_j = 0$.

The rigid firm charges the fixed price p_0. Its profit is $\Pi_{rigid} = p_0 \cdot q_{rigid} - C(q_{rigid})$. As $\tau \to \infty$, $q_{rigid} \to 0$, and:

$$\lim_{\tau \to \infty} \Pi_{rigid}(\tau) = (p_0 \cdot 0) - C(0) = 0$$

The standard firm's profit Π^* must be non-negative (it can always choose $p_j = 0$, sell $q = 0$, and earn $\Pi = 0$). Since its profit is also choked off by the tax, its optimal profit must also converge to 0:

$$\lim_{\tau \to \infty} \Pi^*(\tau) = 0$$

Therefore, the limit of the profit difference is:

$$\lim_{\tau \to \infty} \Delta\Pi_{rigid}(\tau) = \lim_{\tau \to \infty} \Pi_{rigid}(\tau) - \lim_{\tau \to \infty} \Pi^*(\tau) = 0 - 0 = 0$$

In conclusion, we have a continuous function $\Delta\Pi_{rigid}(\tau)$ such that $\Delta\Pi_{rigid}(0) = 0$, $\frac{d}{d\tau}\Delta\Pi_{rigid}(0) > 0$, and $\lim_{\tau \to \infty} \Delta\Pi_{rigid}(\tau) = 0$. A function that starts at 0, increases, and must eventually return to 0 must be non-monotonic. □

Proposition 8 (The Value of Flexibility)**.** The profit advantage of a flexible firm over a rigid firm within the friction model, $\Delta\Pi_{flex,rigid} = \Pi_{flex} - \Pi_{rigid}$, is a strictly increasing function of the tax rate τ.

Proof. Let $V(\bar{p}_\infty, \tau) = \Pi_{max}(\bar{p}_\infty, \tau) - \Pi_{rigid}(\bar{p}_\infty, \tau)$ be the value of flexibility, where Π_{max} is the flexible firm's value function and $\Pi_{rigid} = \Pi(p_0, \bar{p}_\infty, \tau)$. We take the total derivative of V with respect to τ:

$$\frac{dV}{d\tau} = \left(\frac{\partial V}{\partial \bar{p}_\infty} \cdot \frac{d\bar{p}_\infty}{d\tau}\right) + \left(\frac{\partial V}{\partial \tau}\right)$$

We prove that both additive terms are positive.

1. **The Direct Effect ($\frac{\partial V}{\partial \tau}$):** By the Envelope Theorem, $\frac{\partial \Pi_{max}}{\partial \tau} = \frac{\partial \Pi}{\partial \tau}|_{p^*_\infty}$, while $\frac{\partial \Pi_{rigid}}{\partial \tau} = \frac{\partial \Pi}{\partial \tau}|_{p_0}$. Thus:
$$\frac{\partial V}{\partial \tau} = \frac{\partial \Pi}{\partial \tau}\bigg|_{p^*_\infty} - \frac{\partial \Pi}{\partial \tau}\bigg|_{p_0}$$
By Assumption 4(Tax Effect), $\frac{\partial^2 \Pi}{\partial p_j \partial \tau} < 0$, so $\frac{\partial \Pi}{\partial \tau}$ is a decreasing function of p_j. Since $p^*_\infty < p_0$ (from Prop 1), it must be that $\frac{\partial \Pi}{\partial \tau}|_{p^*_\infty} > \frac{\partial \Pi}{\partial \tau}|_{p_0}$. This term is **positive**.

2. **The Environmental Effect ($\frac{\partial V}{\partial \bar{p}_\infty} \cdot \frac{d\bar{p}_\infty}{d\tau}$):** This term is a product of two components:

   - $\frac{d\bar{p}_\infty}{d\tau}$: By Lemma 1 and Assumption 4, the producer price falls with the tax. This derivative is **negative**.
   - $\frac{\partial V}{\partial \bar{p}_\infty} = \frac{\partial \Pi_{max}}{\partial \bar{p}_\infty} - \frac{\partial \Pi_{rigid}}{\partial \bar{p}_\infty} = \frac{\partial \Pi}{\partial \bar{p}}\bigg|_{p^*_\infty} - \frac{\partial \Pi}{\partial \bar{p}}\bigg|_{p_0}$. By Core Assumption 2 (Strategic Complementarity), $\frac{\partial^2 \Pi}{\partial p_j \partial \bar{p}} > 0$, so $\frac{\partial \Pi}{\partial \bar{p}}$ is an increasing function of p_j. Since $p^*_\infty < p_0$, it must be that $\frac{\partial \Pi}{\partial \bar{p}}|_{p^*_\infty} < \frac{\partial \Pi}{\partial \bar{p}}|_{p_0}$. This derivative is **negative**.

   The product of two negative terms is **positive**.

Since $\frac{dV}{d\tau}$ is the sum of two strictly positive terms, it is strictly positive. □

Proposition 9 (Rigidity Bound)**.** Define the profit difference of a rigid firm relative to a standard firm as
$$\Delta\Pi(\alpha_1) = \Pi_{rigid}(\alpha_1) - \Pi^*,$$
where α_1 denotes the share of rigid firms in the market. Suppose $\Delta\Pi(\alpha_1)$ is strictly increasing in α_1 with $\Delta\Pi(0) < 0$.

Then a rigid firm is better off than a standard firm if and only if $\alpha_1 > \alpha_1^*$, where the unique threshold $\alpha_1^* \in (0, 1)$ exists precisely when $\Delta\Pi(1) > 0$.

Proof. By continuity of the underlying demand and pricing functions, $\Delta\Pi(\alpha_1)$ is continuous. Moreover, only the market average price responds to α_1, so
$$\frac{d\Delta\Pi}{d\alpha_1} = \frac{d\Pi_{rigid}}{d\alpha_1} = \frac{\partial \Pi}{\partial \bar{p}_\infty} \cdot \frac{d\bar{p}_\infty}{d\alpha_1}.$$
Assumption 3 implies $\frac{\partial \Pi}{\partial \bar{p}_\infty} > 0$, and Proposition 1 implies $\frac{d\bar{p}_\infty}{d\alpha_1} > 0$. Hence $\frac{d\Delta\Pi}{d\alpha_1} > 0$, so $\Delta\Pi(\alpha_1)$ is strictly increasing.

At $\alpha_1 = 0$, $\bar{p}_\infty = p^*$, and p^* is the unique maximizer of $\Pi(\cdot, p^*, \tau)$. Since $p_0 \neq p^*$,

$$\Delta\Pi(0) = \Pi(p_0, p^*, \tau) - \Pi(p^*, p^*, \tau) < 0.$$

We have a continuous, strictly increasing function with $\Delta\Pi(0) < 0$. By the Intermediate Value Theorem, a unique $\alpha_1^* \in (0, 1)$ such that $\Delta\Pi(\alpha_1^*) = 0$ exists if and only if $\Delta\Pi(1) > 0$.

If $\Pi_{rigid}(1) > \Pi^*$, such a root exists and strict monotonicity implies that

$$\Delta\Pi(\alpha_1) > 0 \quad \Longleftrightarrow \quad \alpha_1 > \alpha_1^*.$$

□

Proposition 10 (Effect of Tax on CTB)**.** The difference in the consumer tax burden between the friction model and the standard model, $\Delta CTB = CTB_{friction} - CTB_{standard}$, is a strictly increasing function of the tax rate τ.

Proof. Define the consumer burden gap as a function of the tax rate. $\Delta CTB(\tau) = \overline{p}_{c,\infty} - p_c^* = \bar{p}_\infty(1 + \tau) - p^*(1 + \tau)$. Let $G_p(\tau) = \bar{p}_\infty(\tau) - p^*(\tau)$. Then $\Delta CTB(\tau) = (1 + \tau)G_p(\tau)$.

Using the product rule:

$$\frac{d(\Delta CTB)}{d\tau} = \underbrace{G_p(\tau)}_{>0 \text{ from Prop. } 1 \text{ and } 3} + \underbrace{(1+\tau)\frac{dG_p}{d\tau}}_{\text{by Lemma } 1} > 0$$

□

Proposition 11 (The Effect of Rigidity on the CTB)**.** The difference in the consumer tax burden, $\Delta CTB = CTB_{friction} - CTB_{standard}$, is a strictly increasing function of the proportion of rigid firms, α_1.

Proof. Define the consumer burden gap as a function of the rigidity. $\Delta CTB(\alpha_1) = \overline{p}_{c,\infty}(\alpha_1) - p_c^*$. Differentiating with respect to α_1 (p_c^* is a constant w.r.t. α_1):

$$\frac{d(\Delta CTB)}{d\alpha_1} = \frac{d\overline{p}_{c,\infty}}{d\alpha_1} = \frac{d}{d\alpha_1}(\bar{p}_\infty(1 + \tau)) = (1 + \tau)\frac{d\bar{p}_\infty}{d\alpha_1}$$

which is positive as proved in Proposition 1 □

C.3 Dynamics

Proposition 12 (Dynamic Tax Incidence)**.** Following the imposition of the tax at $t = 0$, and assuming a stable, monotonic price convergence (i.e., $\overline{p}_t < \overline{p}_{t-1}$ for all $t \geq 1$), the Consumer Tax Burden (CTB) is a monotonically decreasing function of time, while the Producer Tax Burden (PTB) is a monotonically increasing function of time.

Proof. The burdens at any time t are defined relative to the initial price p_0 and the average price at time t, $\overline{p}_t$ can be defined as follows.

$$PTB(0) = p_0 - p_0 \tag{22}$$

$$CTB(0) = p_0(1 + \tau) - p_0 = \tau p_0 \tag{23}$$

At the moment the tax is imposed, no prices have changed, so $\overline{p}_0 = p_0$.

$$PTB(t) = p_0 - \overline{p}_t$$
$$CTB(t) = \overline{p}_t(1 + \tau) - p_0$$

The Law of Motion governs the adjustment of $\overline{p}_t$ from its initial state p_0 to its new, lower long-run state $\overline{p}_\infty$ (where $\overline{p}_\infty < p_0$ as established in Proposition 1). Additionally, we assume the price path does not oscillate and is a strictly decreasing sequence:

$$\overline{p}_t < \overline{p}_{t-1} \quad \text{for } t \geq 1$$

Next, we compare tax burdens between periods t and $t - 1$. Comparison of $PTB(t)$ to $PTB(t - 1)$ yields,

$$PTB(t) = p_0 - \overline{p}_t$$

Since $\overline{p}_t < \overline{p}_{t-1}$, it follows that $-\overline{p}_t > -\overline{p}_{t-1}$.

$$p_0 - \overline{p}_t > p_0 - \overline{p}_{t-1} \implies PTB(t) > PTB(t - 1)$$

The Producer Tax Burden is monotonically increasing from its minimum of 0.

Comparison of $CTB(t)$ to $CTB(t - 1)$ yields,

$$CTB(t) = \overline{p}_t(1+\tau) - p_0$$

Since $\overline{p}_t < \overline{p}_{t-1}$, it follows that $\overline{p}_t(1+\tau) < \overline{p}_{t-1}(1+\tau)$.

$$\overline{p}_t(1+\tau) - p_0 < \overline{p}_{t-1}(1+\tau) - p_0 \implies CTB(t) < CTB(t-1)$$

The Consumer Tax Burden is monotonically decreasing from its maximum of τp_0. Therefore, we can conclude that consumers bear the maximum burden at $t = 0$, and this burden is gradually shifted to producers as time progresses and firms adjust their prices.

$\square$

Proposition 13 (The Source of Dynamic Persistence)**.** The model's dynamic "memory" (i.e., its dependence on prices before $t-1$) is generated exclusively by the presence of partially flexible (Type 2) firms. In the absence of Type 2 firms ($\alpha_2 = 0$), the model's dynamics simplify to a first-order process.

Proof. The average price $\overline{p}_t$ is the weighted average of all firm types:

$$\overline{p}_t = \alpha_1 p_0 + \alpha_2 \overline{p}_{2,t} + \alpha_3 p_t^*$$

where $p_t^* = p^*(\overline{p}_{t-1}, \tau)$ is the optimal price. $\overline{p}_{2,t}$ is the average price of Type 2 (Calvo) firms. By definition, a fraction μ_2 of them update to p_t^*, while the remaining $(1-\mu_2)$ fraction keep their price from $t-1$.

$$\overline{p}_{2,t} = (1-\mu_2)\overline{p}_{2,t-1} + \mu_2 p_t^*$$

This recursive definition means that $\overline{p}_{2,t}$ is a weighted average of *all past optimal prices* $(p_t^*, p_{t-1}^*, p_{t-2}^*...)$. This is what creates the higher-order dynamics (persistence). If there are no Calvo firms, the Law of Motion becomes:

$$\overline{p}_t = \alpha_1 p_0 + \alpha_3 p_t^* \quad \text{(where } \alpha_1 + \alpha_3 = 1\text{)}$$

Substituting the best-response function:

$$\overline{p}_t = \alpha_1 p_0 + \alpha_3 p^*(\overline{p}_{t-1}, \tau)$$

This is a first-order difference equation. $\overline{p}_t$ depends *only* on $\overline{p}_{t-1}$. All "memory" of $\overline{p}_{t-2}$ and earlier is lost.

$\square$

Proposition 14 (Rigidity and Convergence Speed)**.** A higher proportion of rigid firms (α_1) slows the market's convergence to its long-run equilibrium.

Proof. The market's dynamic adjustment is driven by the mass of non-rigid firms (Types 2 and 3), which has a total measure of $(1 - \alpha_1)$. These are the only firms that can update their prices in response to the tax and move the average price $\overline{p}_t$ toward its new steady state $\overline{p}_\infty$.

A higher proportion of rigid firms α_1 mechanically reduces the mass of adjusting firms $(1 - \alpha_1)$.

This makes the aggregate price $\overline{p}_t$ less responsive to any deviation from the steady state. In the Law of Motion (Equation 7), the coefficients on the dynamic terms (e.g., $p^*(\overline{p}_{t-1}, \tau)$) are all scaled by $(\alpha_2 + \alpha_3) = (1 - \alpha_1)$. A larger α_1 reduces the magnitude of these terms, dampening the system's response.

A less responsive system, by definition, converges more slowly to its long-run equilibrium following a shock. $\square$

Proposition 15 (Consumer Price Overshooting)**.** Following the tax, the average consumer price $\overline{p}_t^c$ overshoots its new long-run equilibrium. It jumps to its highest point at $t = 0$ and then (assuming monotonic convergence) decreases to its final steady state, $\overline{p}_{c,\infty}$.

Proof. The proof consists of two parts. We first compare the start and end points of the price path, and then characterize the path between them.

Step 1: The consumer price at a fixed time t is $\overline{p}_t^c = \overline{p}_t(1 + \tau)$.

At $t = 0$, no prices have adjusted, so $\overline{p}_0 = p_0$. The consumer price instantly becomes:

$$\overline{p}_0^c = p_0(1 + \tau)$$

In the long-run steady state ($t \to \infty$), the producer price is $\overline{p}_\infty$. The consumer price is:

$$\overline{p}_{c,\infty} = \overline{p}_\infty(1 + \tau)$$

From the long-run aggregation equation (Eq. 12), $\overline{p}_\infty$ is a weighted average of p_0 and p^*_∞. From the proof of Proposition 1, we know $p^*_\infty < p_0$. Therefore, it must be that $\overline{p}_\infty < p_0$.

Since $\overline{p}_\infty < p_0$, it follows that the final consumer price is strictly lower than the initial consumer price:

$$\overline{p}_{c,\infty} < \overline{p}_0^c$$

This proves the initial consumer price is strictly higher than the final long-run consumer price, meaning the price must, in net, fall to reach its new equilibrium.

Step 2: To prove the path is monotonically decreasing, we must characterize the sequence $\{\overline{p}_t^c\}_{t=1}^{\infty}$. We make the standard assumption that the parameters of the model (A.1-A.4) are such that the dynamic system converges to its steady state monotonically, without overshooting or oscillating. This means the producer price path is a strictly decreasing sequence:

$$\overline{p}_t < \overline{p}_{t-1} \quad \text{for all } t \geq 1$$

Given this, it follows directly that the consumer price path is also monotonically decreasing for all $t \geq 1$:

$$\overline{p}_t(1+\tau) < \overline{p}_{t-1}(1+\tau) \implies \overline{p}_t^c < \overline{p}_{t-1}^c$$

Thus, the average consumer price jumps to its highest point at $t = 0$ and then monotonically decreases to its final steady state. $\square$